



TUBAF
Die Ressourcenuniversität.
Seit 1765.



Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

To the Faculty of Mechanical, Process and Energy Engineering
of the Technische Universität Bergakademie Freiberg

is submitted this

Master Thesis

to attain the academic degree of

Master of Science
(M. Sc.)

Course of Study: Computational Materials Science

Submitted by:

Pruthviraja Reddy Vandavagali

Matriculation number:

68865

1st Examiner (Supervisor):

Prof. Dipl.-Ing. Björn Kiefer, Ph.D.

2nd Examiner (Reviewer):

Dr.-Ing. Stephan Roth

Institute:

Institute of Mechanics and Fluid Dynamics (IMFD)

Chair:

Chair of Applied Mechanics – Solid Mechanics

Freiberg, 22. August 2026

Abstract

Phase-field fracture methods in commercial finite element software via user subroutines (UEL/UMAT) provide a variational framework for regularized crack initiation, propagation, and branching. Resolving the material length scale l_0 requires fine spatial discretizations ($h \leq l_0/2$), making uniform fine meshes computationally demanding for large fracture analyses. This thesis investigates error-indicator-guided mesh refinement and nonmatching multi-field state transfer for phase-field fracture simulations in Abaqus.

The methodology begins with baseline verification of the staggered phase-field formulation of Molnar and Gravouil [18], examining uniform mesh convergence (H_0 , H_1 , H_2 -PUB), reaction force histories, and integration-point irreversibility. To alleviate uniform mesh costs, an error-guided pre-refinement workflow is developed using the Abaqus recovery-based stress discretization error indicator (MISESeri) grounded in the Zienkiewicz–Zhu energy norm [24, 25], following Pandey and Kumar [21]. An elastic pre-analysis identifies stress concentrators, enabling offline generation of graded meshes with localized refinement in the prospective fracture zone.

To establish the prerequisites for multi-stage adaptive simulation, a multi-field state-transfer framework is formulated across nonmatching meshes. The transfer pipeline combines shape-function interpolation for primary displacement and phase fields with centroid-based mapping and positivity/monotonicity clamping for the crack-driving history field $\mathcal{H}$. State transfer is verified across same-mesh restarts, multi-resolution grids, and a synthetic single-facet crack-tip topology cut with post-peak continuation. Karush–Kuhn–Tucker multiplier analyses demonstrate that fixed active sets held frozen without dynamic updating are inadmissible under advancing load, supporting the history-field formulation adopted in the present restart framework.

The framework is validated on a Mode-II shear phase-field benchmark ($G_c = 2.7 \times 10^{-3}$ kN/mm, $l_0 = 0.015$ mm). Censoring-corrected uniform studies (H_0 with 3,930 elements, H_1 with 12,064 elements, H_2 with 33,852 elements) establish the common pre-peak verification window ($0 \leq u_1 \leq 0.009250$ mm) and reveal that spatial crack paths remain mesh-sensitive ($5.44 \mu\text{m}$ Hausdorff distance, exceeding $h_{H_2}/2 = 3.75 \mu\text{m}$ for the fine $h_{H_2} = 7.5 \mu\text{m}$ grid) despite tight global force convergence (0.518%). Two production error-guided refined models are validated: a Minimum-Maximum candidate (MM, 2,206 elements) and a Uniform-Error candidate (PK5, 4,894 elements). Over the common pre-peak window, both candidates achieve high fidelity against the fine H_1 reference, exhibiting normalized L_2 force errors of 1.4249% (MM) and 1.1467% (PK5), relative work errors of 0.8134% and 0.5751%, and stiffness agreement within 0.2425%. Discrete frame audits confirm zero violations of phase bounds ($0 \leq d \leq 1$), history positivity ($\mathcal{H} \geq 0$), and temporal irreversibility ($\Delta d \geq 0, \Delta \mathcal{H} \geq 0$). Using exact scheduler CPU accounting, the refined models yield diagnostic ratios of $12.16\times$ (MM) and $5.54\times$ (PK5) relative to the censored H_2 baseline. Crucially, this work distinguishes between offline error-guided pre-refinement on fixed optimized meshes and dynamic in-analysis adaptive remeshing (a documented future extension).

Declaration on the Use of AI Tools

ChatGPT, Google Gemini, and GitHub Copilot were used as supporting tools during the preparation of this thesis. The author remains fully responsible for the content and results presented.

Contents

Abstract	II
Declaration on the Use of AI Tools	III
1. Introduction, Theoretical Foundations, and Literature Review	1
1.1. Motivation and Problem Statement	1
1.2. Variational Formulation of Phase-Field Brittle Fracture	2
1.2.1. Regularized Energy Functional	2
1.2.2. Strain Energy Decomposition and Constitutive Degradation	2
1.2.3. Irreversibility and the Crack-Driving History Field	3
1.2.4. Coupled Governing Equations and Staggered Solution Scheme	4
1.3. Implementation Architecture in Commercial FE Software	4
1.3.1. Layered User-Element (UEL / UMAT) Architecture	4
1.3.2. State Variable Mapping Contract	5
1.4. Recovery-Based Error Estimation and Pre-Refinement	5
1.4.1. The Zienkiewicz–Zhu Energy Norm Error Estimator	5
1.4.2. Recovered-Stress Interpretation of Abaqus MISESeri	6
1.4.3. The Pandey–Kumar Pre-Refinement Methodology	6
1.5. Multi-Field State Transfer across Nonmatching Meshes	6
1.5.1. Primary Field Transfer (Nodal Interpolation)	6
1.5.2. Secondary History Field Transfer (Gauss-Point Mapping)	7
1.6. Scope of Work and Thesis Organization	7
1.6.1. Distinction of Methodological Scope	7
1.6.2. Thesis Organization	8
2. Baseline Implementation and Verification	9
2.1. Introduction and Objectives	9
2.2. Benchmark Problem: Single-Edge-Notched Tension Test	9
2.2.1. Geometry and Boundary Conditions	9
2.2.2. Material and Regularization Parameters	10
2.2.3. Loading Protocol and Incrementation	10
2.3. Abaqus Staggered Implementation Verification	10
2.4. Uniform Mesh Discretization Sensitivity ($l_0 = 0.015$ mm)	11
2.5. Integration-Point State Variable and Irreversibility Audit	13
2.6. Chapter Summary	13
3. Error-Indicator-Guided Mesh Refinement Methodology	14
3.1. Introduction and Theoretical Motivation	14
3.2. The Zienkiewicz–Zhu Error Recovery Indicator in Abaqus	14
3.2.1. Stress Discretization Error Indicator (MISESeri)	14
3.2.2. Adopted Error Normalization and Sizing Procedure	15

3.3. Offline Pre-Refinement Workflow	15
3.4. Mode-I Verification and Computational Performance	16
3.4.1. Reaction Force and Global Response Comparison	16
3.4.2. Assessment and Methodological Limitations	16
4. Nonmatching State Transfer and Restart Methodology	18
4.1. Introduction and Problem Formulation	18
4.2. Mathematical Formulation of Multi-Field Transfer Operators	18
4.2.1. Primary Field Projection (Nodal Quantities)	18
4.2.2. Secondary History Field Mapping (Gauss-Point History)	19
4.3. The Four-Stage Restart Protocol	19
4.4. Active-Set Formulations and KKT Admissibility Analysis	20
4.4.1. Active-Set Freeze Formulation (Stage D3D)	20
4.4.2. Forensic Analysis of Frozen Active Sets under Restart	20
4.4.3. Offline KKT Correction (Candidate D3D-A1)	21
4.5. Chapter Summary	22
5. Multi-Resolution and Topology-Transfer Benchmarks	23
5.1. Introduction and Objectives	23
5.2. Multi-Resolution Refinement and Coarsening Transfer	23
5.2.1. Benchmark Configuration and Donor Checkpoint	23
5.2.2. Slit-Barrier Isolation and Invariant Preservation	23
5.3. Mode-II Pure Shear Phase-Field Fracture Benchmark	24
5.3.1. Problem Formulation and Boundary Conditions	24
5.3.2. Model Lineage and Disambiguation	24
5.4. Uniform Grid Refinement and Censoring Analysis	25
5.4.1. Censored Peak Force and Common-Window Formulation	25
5.4.2. Key Findings from Uniform Refinement	25
5.5. Synthetic Crack-Tip Topology-Transfer Validation	26
5.5.1. Synthetic Single-Facet Extension Formulation	26
5.5.2. Validation Results and Post-Peak Trajectory	27
5.6. Chapter Summary	27
6. High-Performance Computing Implementation and Shared-State Analysis	28
6.1. Introduction and Parallelization Challenges in Phase-Field UELs	28
6.2. Solver Environment and Toolchain Architecture	28
6.3. Abaqus UEXTERNALDB Common-Block Memory Architecture	29
6.3.1. Inter-Subroutine Shared State Communication	29
6.3.2. Shared-State Safety in Multi-Threaded Execution	29
6.4. Thread-Safety and Parallelization Boundary Audit	29
6.4.1. Controlled Rank and Thread Indexing	29
6.4.2. Formal Parallel Safety Claims and Boundaries	30
6.5. Numerical Determinism and Reproducibility	30
6.6. Chapter Summary	30
7. Production Error-Guided Locally Refined Fracture Validation	31
7.1. Introduction and Objectives	31
7.2. Production Mesh Candidates and Execution Provenance	31

7.3.	Execution Validation and State-Admissibility Verification	32
7.3.1.	Execution Verification	32
7.3.2.	State Admissibility and Irreversibility Audits across Saved ODB Frames	32
7.4.	Quantitative Accuracy in the Common Pre-Peak Domain	33
7.4.1.	Global Force-Displacement Response ($0 \leq u_1 \leq 0.009250$ mm)	33
7.4.2.	Reconciliation of Pointwise Force Discrepancies vs Global L_2 Error	33
7.4.3.	Damage Initiation Observations ($d \geq 0.5$)	34
7.4.4.	Domain-B Continuation Diagnostics ($0.009250 < u_1 \leq 0.010000$ mm)	34
7.5.	Computational Efficiency and Runtime Scaling	35
7.6.	Energy Bookkeeping and Methodological Scope	35
7.6.1.	External Work and Internal Dissipation Accounting	35
7.6.2.	Scope Boundary: Pre-Refinement vs Online Adaptivity	36
7.7.	Chapter Summary	36
8.	Synthesis, Decision Framework, and Conclusions	38
8.1.	Synthesis of Validated Research Outcomes	38
8.2.	Engineering Decision Tree for Phase-Field Mesh Adaptation	38
8.3.	Methodological and Scientific Limitations	39
8.4.	Future Research Roadmap	40
8.5.	Conclusions	40
A.	Reproducibility Tables, Job Execution Provenance, and Diagnostic Logs	41
A.1.	Master High-Performance Computing Job Ledger	41
A.2.	Mode-I Uniform-Convergence Provenance	41
A.3.	Cryptographic Checksums and Package Provenance	42
A.4.	Historical Solver Diagnostic Codes and Failure Mode Catalog	42
A.5.	Operational HPC Monitoring Protocols	42

List of Figures

2.1. Vertical reaction force vs. displacement for reconstructed Molnar benchmark.	11
2.2. Force-displacement convergence across uniform mesh levels.	12
3.1. Force-displacement comparison of pre-refined mesh against uniform references.	16
4.1. Overview of the four-stage multi-field state transfer and restart workflow in Abaqus.	20
4.2. Evolution of cumulative active-set Lagrange multiplier violations under continuation loading.	21
7.1. Mode-II shear reaction force vs. prescribed displacement curves.	34
7.2. Pointwise relative force discrepancy of locally refined candidates vs. H_1 . . .	35
7.3. Mesh size reduction and CPU execution time comparison.	36

List of Tables

2.1. Reconstructed Molnar SENT Benchmark Quantitative Metrics and Validation Audit.	10
2.2. Discretization Statistics and Convergence Metrics for Uniform Meshes ($l_0 = 0.015$ mm).	12
3.1. Quantitative Accuracy and Computational Cost Comparison: Pre-Refined vs Uniform Meshes.	16
5.1. Model Lineage and Canonical Role Disambiguation for Mode-II Simulations.	25
5.2. Mode-II FRACFIX Uniform Grid Refinement Comparison and Common-Window Metrics.	26
7.1. Discretization Statistics and Computational Resource Usage in Production Validation.	32
7.2. State-Admissibility and Irreversibility Verification across Saved ODB Field Frames.	32
7.3. Domain-A Accuracy Metrics against H_1 Reference ($0 \leq u_1 \leq 0.009250$ mm).	33
7.4. Damage Initiation Observations ($d \geq 0.5$) across Saved Field Frames. . . .	34
8.1. Comprehensive Synthesis of Thesis Research Outcomes and Validation Boundaries.	38
A.1. Authoritative HPC Simulation Job Ledger across All Research Stages. . . .	41
A.2. Traceability of the Mode-I Uniform-Convergence Data.	42
A.3. Cryptographic SHA-256 Checksums for Production Executable Artifacts (Input Deck, User Subroutine, PBS Directives, and Package Manifest).	42
A.4. Catalog of Technical Failure Modes and Engineering Resolutions.	43

1. Introduction, Theoretical Foundations, and Literature Review

1.1. Motivation and Problem Statement

Predicting crack initiation, propagation, and complex branching patterns in brittle and quasi-brittle solids is a central challenge in computational solid mechanics and materials engineering [3]. Classical linear elastic fracture mechanics (LEFM), initiated by Griffith [11] and extended by Irwin [13], provides rigorous criteria for crack growth based on energy release rates and stress intensity factors. However, standard discrete fracture models—such as cohesive zone models or the extended finite element method (XFEM)—require explicit tracking of geometric discontinuities, geometric enrichment, or complicated remeshing procedures when handling arbitrary three-dimensional crack paths, crack coalescence, or spontaneous branching.

The **variational phase-field approach to fracture**, pioneered by Francfort and Marigo [10] and computationally realized by Bourdin, Francfort, and Marigo [4, 5], overcomes these difficulties by regularizing sharp crack surfaces into a continuous diffusive damage corridor governed by a continuous scalar field $d \in [0, 1]$. In this framework, crack topology is implicitly tracked by the phase field, and the crack path emerges naturally from energy minimization without requiring ad-hoc tracking criteria or discontinuous displacement jumps [15].

Despite its mathematical elegance, the phase-field method imposes severe computational demands. To accurately capture the steep gradients of the phase field across the regularized fracture surface, the finite element mesh size h must sufficiently resolve the internal material length scale l_0 . Standard convergence studies dictate that at least two to five elements must span the half-width of the diffuse zone ($h \leq l_0/2$). For fine structural features or thin specimens where l_0 is on the order of micrometers, a uniform discretization of the entire domain leads to massive system dimensions (10^5 to 10^7 degrees of freedom), making multi-step or full post-peak fracture simulations computationally prohibitive [12, 22].

To mitigate this computational bottleneck, adaptive mesh refinement strategies have become essential [2, 20]. In commercial finite element suites such as Abaqus, user-defined element subroutines (UELs) and user material subroutines (UMATs) are commonly employed to solve coupled phase-field systems [8, 18, 19]. However, integrating Abaqus' built-in adaptive remeshing tools with multi-field UEL formulations presents non-trivial challenges:

1. Standard built-in error indicators (such as the recovery-based stress error indicator MISESRI) operate primarily on native continuum elements rather than custom multi-field user elements.
2. Commercial solvers lack native state-mapping routines for custom UEL integration-point history variables ($\mathcal{H}$) across nonmatching grids.

3. Dynamic crack growth requires rigorous preservation of thermodynamic bounds ($0 \leq d \leq 1$) and irreversible crack growth ($\dot{d} \geq 0$) when transferring state data between meshes.

This thesis addresses these challenges by investigating the application of Abaqus built-in error indicators, error-guided local mesh refinement, and nonmatching multi-field state transfer to phase-field fracture simulations.

1.2. Variational Formulation of Phase-Field Brittle Fracture

1.2.1. Regularized Energy Functional

Consider an elastic body occupying a domain $\Omega \subset \mathbb{R}^\delta$ ($\delta \in \{2, 3\}$) with external boundary $\partial\Omega = \Gamma_u \cup \Gamma_t$ ($\Gamma_u \cap \Gamma_t = \emptyset$). Following the variational formulation of Francfort and Marigo [10] and Bourdin et al. [4], the total potential energy of the fracturing solid is expressed as the sum of the elastic strain energy and the fracture surface energy:

$$\Pi(\mathbf{u}, d) = \int_{\Omega} \psi(\boldsymbol{\varepsilon}(\mathbf{u}), d) \, d\Omega + G_c \int_{\Omega} \gamma(d, \nabla d) \, d\Omega - \int_{\Omega} \mathbf{b} \cdot \mathbf{u} \, d\Omega - \int_{\Gamma_t} \mathbf{t} \cdot \mathbf{u} \, d\Gamma, \quad (1.1)$$

where $\mathbf{u}(\mathbf{x})$ is the displacement field, $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top)$ is the small-strain tensor, G_c is the critical energy release rate (fracture toughness), $\mathbf{b}$ denotes body forces, $\mathbf{t}$ represents prescribed surface tractions, and $\gamma(d, \nabla d)$ is the crack surface density function.

The continuous scalar phase field $d(\mathbf{x}, t) \in [0, 1]$ serves as an interpolation parameter between the undamaged elastic state ($d = 0$) and the completely broken fracture state ($d = 1$). In this work, we adopt the standard second-order regularized AT2 crack surface density function [4, 17]:

$$\gamma(d, \nabla d) = \frac{1}{2l_0} d^2 + \frac{l_0}{2} |\nabla d|^2, \quad (1.2)$$

where $l_0 > 0$ is the regularization parameter possessing the dimension of length, governing the spatial width of the diffuse crack band. In the asymptotic limit $l_0 \rightarrow 0$, the regularized crack surface energy $\int_{\Omega} G_c \gamma(d, \nabla d) \, d\Omega$ Γ -converges to the Griffith discrete crack surface energy $G_c \mathcal{H}^{\delta-1}(\Gamma_c)$ [5].

1.2.2. Strain Energy Decomposition and Constitutive Degradation

To prevent crack propagation and unphysical material interpenetration under compressive stress states, the undamaged elastic strain energy density $\psi_0(\boldsymbol{\varepsilon})$ is decomposed into positive (tensile) and negative (compressive) parts:

$$\psi_0(\boldsymbol{\varepsilon}) = \psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}). \quad (1.3)$$

Constitutive damage degradation acts exclusively on the tensile strain energy component $\psi_+(\boldsymbol{\varepsilon})$ through a monotonic quadratic degradation function $g(d)$:

$$\psi(\boldsymbol{\varepsilon}, d) = g(d)\psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}) = [(1-d)^2 + k_{\text{res}}] \psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}), \quad (1.4)$$

where $k_{\text{res}} \ll 1$ (typically $k_{\text{res}} = 10^{-7}$) is a positive residual numerical stiffness parameter that ensures positive-definiteness of the mechanical tangent operator in fully broken regions ($d \rightarrow 1$).

In this thesis, we utilize the spectral strain decomposition proposed by Miehe, Welschinger, and Hofacker [17]:

$$\psi_+(\boldsymbol{\varepsilon}) = \frac{1}{2}\lambda\langle\text{tr}(\boldsymbol{\varepsilon})\rangle_+^2 + \mu\text{tr}(\boldsymbol{\varepsilon}_+^2), \quad (1.5)$$

$$\psi_-(\boldsymbol{\varepsilon}) = \frac{1}{2}\lambda\langle\text{tr}(\boldsymbol{\varepsilon})\rangle_-^2 + \mu\text{tr}(\boldsymbol{\varepsilon}_-^2), \quad (1.6)$$

where λ and μ are the standard Lamé elastic constants, $\langle x \rangle_{\pm} = \frac{1}{2}(x \pm |x|)$ denote Macaulay brackets, and the positive/negative strain tensors are constructed from the spectral decomposition of the strain tensor $\boldsymbol{\varepsilon} = \sum_{i=1}^{\delta} \varepsilon_i \mathbf{n}_i \otimes \mathbf{n}_i$:

$$\boldsymbol{\varepsilon}_{\pm} = \sum_{i=1}^{\delta} \langle \varepsilon_i \rangle_{\pm} \mathbf{n}_i \otimes \mathbf{n}_i. \quad (1.7)$$

The Cauchy stress tensor $\boldsymbol{\sigma}$ is obtained by thermodynamic differentiation with respect to strain:

$$\boldsymbol{\sigma} = \frac{\partial \psi(\boldsymbol{\varepsilon}, d)}{\partial \boldsymbol{\varepsilon}} = g(d)\boldsymbol{\sigma}_+ + \boldsymbol{\sigma}_-, \quad (1.8)$$

where $\boldsymbol{\sigma}_+ = \lambda\langle\text{tr}(\boldsymbol{\varepsilon})\rangle_+ \mathbf{I} + 2\mu\boldsymbol{\varepsilon}_+$ and $\boldsymbol{\sigma}_- = \lambda\langle\text{tr}(\boldsymbol{\varepsilon})\rangle_- \mathbf{I} + 2\mu\boldsymbol{\varepsilon}_-$.

1.2.3. Irreversibility and the Crack-Driving History Field

Thermodynamic consistency demands that crack healing is physically prohibited, imposing the local irreversibility constraint:

$$\dot{d}(\mathbf{x}, t) \geq 0 \quad \forall \mathbf{x} \in \Omega. \quad (1.9)$$

Following Miehe et al. [16], this constraint is enforced algorithmically by introducing an internal strain-history variable $\mathcal{H}(\mathbf{x}, t)$, which records the maximum positive tensile strain energy density attained over the deformation history up to current time t :

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \in [0, t]} \psi_+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)). \quad (1.10)$$

At each material integration point, the discrete update between increments n and $n + 1$ is evaluated as:

$$\mathcal{H}_{n+1} = \max(\mathcal{H}_n, \psi_+(\boldsymbol{\varepsilon}_{n+1})), \quad (1.11)$$

guaranteeing pointwise monotonicity $\mathcal{H}_{n+1} \geq \mathcal{H}_n \geq 0$ unconditionally. In the adopted model unit system (kN, mm), $\mathcal{H}$ is evaluated in consistent model energy-density units ($\text{kN}/\text{mm}^2 = \text{kN} \cdot \text{mm}/\text{mm}^3 = \text{GPa}$, equivalent to 10^3 MPa).

1.2.4. Coupled Governing Equations and Staggered Solution Scheme

Stationarity of the regularized energy functional (1.1) with respect to admissible variations $\delta \mathbf{u}$ and δd yields the coupled Euler–Lagrange boundary value problem:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad \text{in } \Omega, \quad (1.12)$$

$$G_c l_0 \nabla^2 d - \left(\frac{G_c}{l_0} + 2\mathcal{H} \right) d + 2\mathcal{H} = 0 \quad \text{in } \Omega, \quad (1.13)$$

subject to the Neumann boundary condition for the phase field:

$$\nabla d \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \quad (1.14)$$

Because the coupled problem is non-convex in $(\mathbf{u}, d)$ simultaneously but separately convex in $\mathbf{u}$ for fixed d , and in d for fixed $\mathbf{u}$, we employ a **staggered (operator-split) solution scheme** [1, 16, 18]. In this scheme, the displacement field $\mathbf{u}_{n+1}$ and phase field d_{n+1} are solved alternately within each time increment, improving numerical robustness during strongly nonlinear fracture evolution.

1.3. Implementation Architecture in Commercial FE Software

1.3.1. Layered User-Element (UEL / UMAT) Architecture

To implement the staggered phase-field framework within Abaqus/Standard, a multi-layer user-defined element architecture is utilized, following the methodology of Molnar and Gravouil [18]:

- **Layer 1 (U1 Phase-Field User Element):** Discretizes the regularized phase-field Helmholtz equation (1.13). The scalar phase field d is assigned to degree of freedom 11 (DOF 11). The element stiffness matrix is symmetric and linear:

$$\mathbf{K}_d^e = \int_{\Omega_e} \left[\left(\frac{G_c}{l_0} + 2\mathcal{H} \right) \mathbf{N}_d^T \mathbf{N}_d + G_c l_0 \mathbf{B}_d^T \mathbf{B}_d \right] d\Omega, \quad (1.15)$$

and the right-hand side element residual vector is:

$$\mathbf{F}_d^e = \int_{\Omega_e} 2\mathcal{H} \mathbf{N}_d^T d\Omega - \mathbf{K}_d^e \mathbf{d}^e. \quad (1.16)$$

- **Layer 2 (U2 Mechanical User Element):** Discretizes the degraded momentum balance equation (1.12), solving for nodal displacement degrees of freedom (DOF 1 = u_1 , DOF 2 = u_2). The element tangent stiffness incorporates the degraded Cauchy stress (1.8).
- **Layer 3 (UMAT Companion Visualization Layer):** Superimposed standard continuum elements (e.g., CPE4) sharing identical nodal coordinates with the UEL layers. This layer provides zero mechanical stiffness but exports internal solution-dependent state variables (SDVs) for post-processing and visualization in Abaqus/CAE.

1.3.2. State Variable Mapping Contract

To ensure consistent data exchange between the user elements and visualization layers, the solution-dependent state variable (SDV) mapping contract is structured across the staggered execution sequence as follows:

- **SDV14:** Carried phase field magnitude $d \in [0, 1]$ utilized by the mechanical displacement element layer (U2/U4) for constitutive stiffness degradation.
- **SDV15:** Current solved scalar phase field $d \in [0, 1]$ exported directly from the phase-field Helmholtz solver layer (U1/U3).
- **SDV16:** Maximum crack-driving tensile strain energy density history field $\mathcal{H}$ (stored in consistent model energy-density units: $\text{kN}/\text{mm}^2 = \text{kN} \cdot \text{mm}/\text{mm}^3 = \text{GPa}$, equivalent to 10^3 MPa).
- **SDV1:** Internal phase variable initialized and updated at the onset of the phase-field subroutine call.

This mapping applies consistently across both the Molnar single-notch baseline (`SingleNotch_v2.for`) and the production mixed-topology FRACFIX implementation (`f42_mixed_uel.for`).

1.4. Recovery-Based Error Estimation and Pre-Refinement

1.4.1. The Zienkiewicz–Zhu Energy Norm Error Estimator

To guide mesh refinement toward regions requiring fine spatial resolution without prior knowledge of the crack trajectory, a posteriori error estimators are employed [2]. Following Zienkiewicz and Zhu [24, 25], the spatial discretization error $\mathbf{e}_\sigma = \boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h$ is measured by comparing the discontinuous raw finite element stress field $\boldsymbol{\sigma}_h$ with a continuous, recovered superconvergent stress field $\boldsymbol{\sigma}^*$.

The global energy norm of the discretization error is defined as:

$$\|\mathbf{e}_\sigma\|_E = \left(\int_{\Omega} (\boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h) : \mathbb{C}^{-1} : (\boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h) \, d\Omega \right)^{1/2} = \left(\sum_{e=1}^{N_{\text{elem}}} \|\mathbf{e}_\sigma\|_{E,e}^2 \right)^{1/2}, \quad (1.17)$$

where $\mathbb{C}$ is the fourth-order elasticity tensor and $\|\mathbf{e}_\sigma\|_{E,e}$ is the element-level error contribution. The relative energy norm error η is normalized by the total strain energy U :

$$\eta = \frac{\|\mathbf{e}_\sigma\|_E}{(U + \|\mathbf{e}_\sigma\|_E^2)^{1/2}} \times 100\%. \quad (1.18)$$

1.4.2. Recovered-Stress Interpretation of Abaqus MISESeri

Abaqus provides MISESeri as a recovery-based equivalent Mises stress discretization error indicator [7]. In this thesis, its behavior is interpreted in the context of the Zienkiewicz–Zhu recovered-stress framework through the conceptual element measure

$$\text{MISESeri}_e = \left(\frac{1}{V_e} \int_{\Omega_e} [q(\boldsymbol{\sigma}^*) - q(\boldsymbol{\sigma}_h)]^2 d\Omega \right)^{1/2}, \quad (1.19)$$

where $q(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2} \mathbf{S} : \mathbf{S}}$ is the equivalent Mises stress, $\mathbf{S} = \boldsymbol{\sigma} - \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \mathbf{I}$ is the stress deviator, and V_e is the element volume. Equation (1.19) is an error-estimation interpretation used in this work, not a claim about the proprietary internal implementation of Abaqus. High MISESeri values identify elements where spatial stress gradients are poorly resolved by the current mesh polynomial order and element size h_e .

1.4.3. The Pandey–Kumar Pre-Refinement Methodology

Pandey and Kumar [21] developed a simple, robust implementation for phase-field fracture modeling in Abaqus by decoupling error-indicator extraction from the non-linear fracture simulation:

1. **Linear Elastic Pre-Analysis:** Execute a rapid, coarse-grid elastic pre-analysis up to a small threshold displacement prior to damage initiation ($u \ll u_{\text{init}}$).
2. **Error Indicator Extraction:** Extract the resulting spatial distribution of MISESeri from the pre-analysis output database (.odb).
3. **Remeshing Rule Application:** Construct an Abaqus RemeshingRule targeting a designated solution error (e.g., target error 5%) with specified minimum element size ($h_{\min} \leq l_0/2$) and maximum size ($h_{\max}$).
4. **Pre-Refined Mesh Generation:** Generate an optimized, graded mesh where fine element sizing is concentrated along the stress-concentration and anticipated crack-initiation corridor.
5. **Production Fracture Analysis:** Execute the full, non-linear phase-field fracture simulation on the pre-refined mesh from the virgin state.

1.5. Multi-Field State Transfer across Nonmatching Meshes

1.5.1. Primary Field Transfer (Nodal Interpolation)

Primary field unknowns—namely nodal displacement vectors $\mathbf{u} = (u_1, u_2)^T$ and nodal phase field d —are defined at mesh vertices. For any target node with spatial coordinates $\mathbf{x}_T \in \Omega_T$, geometric spatial search identifies the host donor element $\Omega_{D,e}$ satisfying $\mathbf{x}_T \in \Omega_{D,e}$. The local natural coordinates $(\xi_T, \eta_T) \in [-1, 1]^2$ are inverted via Newton–Raphson:

$$\mathbf{x}_T = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T) \mathbf{x}_{D,a}, \quad (1.20)$$

and the transferred nodal field $\phi_T(\mathbf{x}_T) \in \{\mathbf{u}_T, d_T\}$ is evaluated directly from donor nodal values $\phi_{D,a}$ [6, 9]:

$$\phi_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T) \phi_{D,a}. \quad (1.21)$$

To preserve thermodynamic admissibility, the transferred phase field is strictly clamped:

$$d_T(\mathbf{x}_T) \leftarrow \min(1.0, \max(0.0, d_T(\mathbf{x}_T))). \quad (1.22)$$

1.5.2. Secondary History Field Transfer (Gauss-Point Mapping)

Unlike nodal quantities, the crack-driving history variable $\mathcal{H}$ is an internal Gauss-point quantity defined at quadrature points $\mathbf{x}_{T,g}$. Transferring integration-point history variables requires specialized operators [14, 23]:

1. **Direct Quadrature Interpolation:** Evaluate donor Gauss-point values $\mathcal{H}_{D,i}$ via donor element shape functions or nearest-centroid projection.
2. **Positivity and Floor Clamping:** Because $\mathcal{H}$ represents an energy density, negative values are unphysical. The transferred field is strictly projected:

$$\mathcal{H}_T(\mathbf{x}_{T,g}) \geq 0 \quad \forall g \in \{1, \dots, n_{\text{gp}}\}. \quad (1.23)$$

3. **Local Phase-Consistency Diagnostic:** To verify consistency between transferred damage d_T and driving history $\mathcal{H}_T$, the local/homogeneous relation (obtained when the spatial Laplacian gradient term $G_c l_0 \nabla^2 d$ is neglected) is evaluated as a diagnostic check:

$$\mathcal{H}_T(\mathbf{x}_{T,g}) \approx \frac{G_c}{2l_0} \frac{d_T(\mathbf{x}_{T,g})}{1 - d_T(\mathbf{x}_{T,g})} \quad \text{for elements with } d_T > 0. \quad (1.24)$$

1.6. Scope of Work and Thesis Organization

1.6.1. Distinction of Methodological Scope

To ensure rigorous scientific boundaries, this thesis strictly distinguishes between three related concepts:

- **Offline Error-Guided Pre-Refinement:** Mesh refinement conducted prior to the fracture simulation, guided by linear elastic error indicators (MISESeri) at stress concentrators (Chapters 3 and 7). This approach is fully validated and evaluated on production benchmarks.
- **Nonmatching State Transfer & Multi-Grid Restart:** The mathematical and computational framework for mapping primary and history state variables across nonmatching discretizations (Chapters 4 and 5). This is validated across benchmark transfer checkpoints and synthetic topology cuts as a necessary prerequisite for adaptive workflows.

- **Online In-Analysis Adaptive Remeshing:** Fully automated, runtime remeshing during crack propagation where the mesh continuously updates and transfers state at every time increment. While online dynamic remeshing during crack propagation remains a documented limitation and future research extension, the essential prerequisites—namely error-guided sizing rules, multi-field state transfer, and restart stabilization—are rigorously established.

1.6.2. Thesis Organization

The remainder of this thesis is structured as follows:

- **Chapter 2 (Baseline Implementation and Verification)** details the implementation of the Molnar Mode-I benchmark, uniform mesh convergence (H_0, H_1, H_2 -PUB), and numerical irreversibility verification.
- **Chapter 3 (Error-Indicator-Guided Mesh Refinement Methodology)** investigates Abaqus MISESRI error extraction, adopted pre-refinement sizing rules, and computational savings in Mode-I fracture.
- **Chapter 4 (Nonmatching State Transfer and Restart Methodology)** develops the multi-field transfer operators, Gauss-point history mapping, and KKT active-set admissibility analysis.
- **Chapter 5 (Multi-Resolution and Topology-Transfer Benchmarks)** validates state transfer across multi-resolution refined/coarsened grids and evaluates synthetic crack-tip topology separation.
- **Chapter 6 (High-Performance Computing Implementation and Shared-State Analysis)** examines shared-memory parallelization in Abaqus UELs, UEXTERNALDB common blocks, thread safety, and cluster execution determinism.
- **Chapter 7 (Production Error-Guided Locally Refined Fracture Validation)** presents the final validation on the Mode-II shear benchmark, evaluating production candidates (MM and PK5), Domain-A accuracy, state admissibility, and computational efficiency gains.
- **Chapter 8 (Synthesis, Decision Framework, and Conclusions)** synthesizes the project findings into a practical engineering decision tree, summarizes validated capabilities, and details open limitations.
- **Appendix A (Reproducibility Tables and Execution Ledger)** provides full HPC scheduler accounting, checksums, and execution provenance logs.

2. Baseline Implementation and Verification

2.1. Introduction and Objectives

Prior to introducing error indicators, adaptive mesh refinement, or multi-field state transfer, a rigorous baseline implementation of the phase-field fracture formulation must be verified in the Abaqus finite element environment. The objective of this chapter is threefold:

1. Reconstruct and verify the canonical staggered brittle fracture benchmark of Molnar and Gravouil [18].
2. Evaluate spatial discretization sensitivity across three uniform grid resolutions (H_0, H_1, H_2 -PUB).
3. Perform an in-depth numerical audit of the state variables (SDVs) to assess integration-point irreversibility and history variable monotonicity.

2.2. Benchmark Problem: Single-Edge-Notched Tension Test

2.2.1. Geometry and Boundary Conditions

The single-edge-notched tension (SENT) benchmark represents a standard Mode-I brittle fracture test widely adopted in the phase-field literature [4, 17, 18]. The geometric domain consists of a square specimen $\Omega = [0, L] \times [0, H]$ with dimensions $L = 1.0$ mm and $H = 1.0$ mm under plane strain conditions and unit thickness ($t = 1.0$ mm).

An initial sharp crack of length $a_0 = 0.5$ mm is positioned along the horizontal symmetry line at $y = 0.5$ mm, extending from the left external edge ($x = 0$) to the specimen center ($x = 0.5$ mm). The crack is geometrically modeled via split-node duplicate boundary definitions along the notch flanks ($y = 0.5^\pm$ mm).

The boundary conditions are defined as:

- **Bottom Boundary** ($y = 0$ mm): Fully fixed in both coordinate directions ($u_1 = 0, u_2 = 0$).
- **Top Boundary** ($y = 1.0$ mm): Fixed horizontally ($u_1 = 0$) with monotonically increasing vertical tensile displacement $u_2 = \Delta u$.
- **Lateral Boundaries** ($x = 0$ mm and $x = 1.0$ mm): Traction-free ($\mathbf{t} = \mathbf{0}$).

2.2.2. Material and Regularization Parameters

The isotropic elastic and fracture material properties match the published benchmark specifications [18]:

- Young's modulus: $E = 210.0 \text{ GPa} = 210.0 \text{ kN/mm}^2$
- Poisson's ratio: $\nu = 0.30$
- Critical energy release rate: $G_c = 2.7 \times 10^{-3} \text{ kN/mm} = 2700 \text{ J/m}^2$
- Phase-field length scale: $l_0 = 0.0075 \text{ mm}$ (fine reference) and $l_0 = 0.015 \text{ mm}$ (coarse/medium sensitivity study)
- Residual stiffness parameter: $k_{\text{res}} = 1.0 \times 10^{-7}$

2.2.3. Loading Protocol and Incrementation

Tensile loading is applied in two sequential displacement steps:

- **Step 1 (Elastic Pre-Loading):** Linear ramp from $u_2 = 0.0 \text{ mm}$ to $u_2 = 0.0050 \text{ mm}$ over 500 increments ($\Delta u_2 = 1.0 \times 10^{-5} \text{ mm}$).
- **Step 2 (Fracture Evolution):** Continued displacement from $u_2 = 0.0050 \text{ mm}$ to $u_2 = 0.0067 \text{ mm}$ over 170 increments ($\Delta u_2 = 1.0 \times 10^{-5} \text{ mm}$).

2.3. Abaqus Staggered Implementation Verification

The single-notch candidate model was executed on the HPC cluster using Abaqus 2023 with custom user subroutines `SingleNotch_v2.for`. The finite element mesh comprises 33,852 physical quadrilateral elements ($h_{\min} = 0.0010 \text{ mm}$, giving $h/l_0 = 0.1333$ for $l_0 = 0.0075 \text{ mm}$) and 101,556 layered UEL/UMAT elements across three functional layers.

The vertical reaction force trajectory $RF_2(u_2)$ extracted at the top boundary is depicted in Fig. 2.1. The quantitative results and comparison against published digitization metrics are summarized in Tab. 2.1.

Tab. 2.1.: Reconstructed Molnar SENT Benchmark Quantitative Metrics and Validation Audit.

Metric / Parameter	Reconstructed Model Value	Published / Digitized Reference
Peak Reaction Force ($RF_{2,\max}$)	0.761702 kN	0.715560 kN
Displacement at Peak ($u_{2,\text{peak}}$)	0.006110 mm	0.005868 mm
Relative Peak Force Difference	+6.45%	—
Relative Peak Displacement Difference	+4.13%	—
Pre-Peak RMSE	0.044136 kN	—
Post-Peak RMSE	0.348093 kN	—
Full-Overlap Normalized RMSE	24.57%	—
Crack Extension ($d \geq 0.90$)	0.0525 mm beyond tip	Fully horizontal path
Crack Extension ($d \geq 0.95$)	0.0505 mm beyond tip	Fully horizontal path

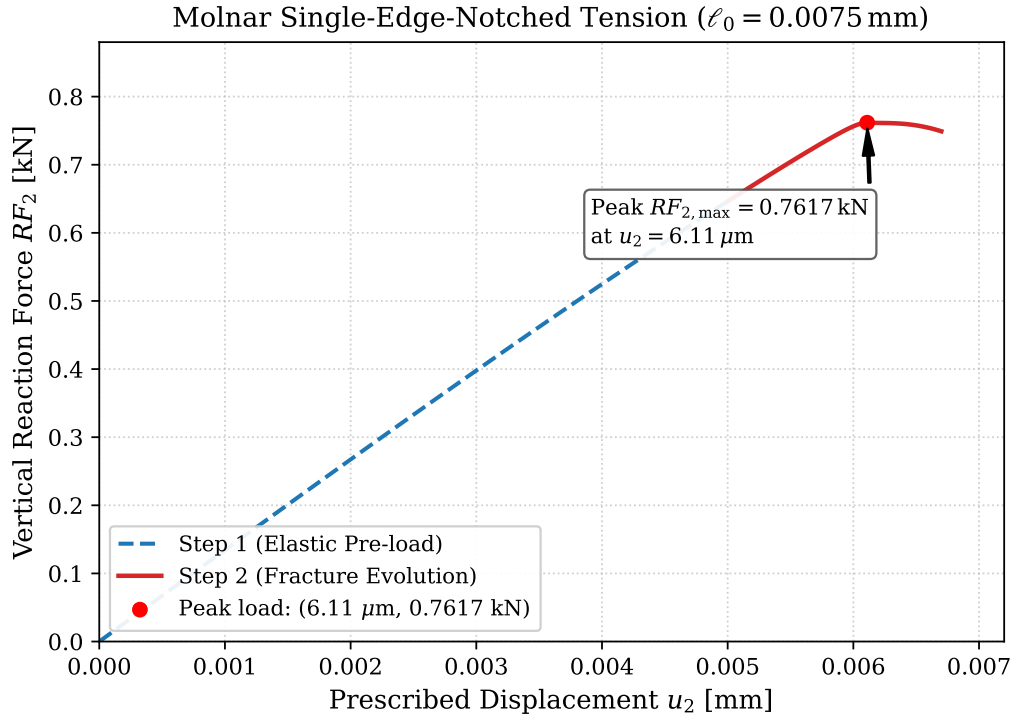


Fig. 2.1.: Vertical reaction force (RF_2 in kN) versus prescribed displacement (u_2 in mm) for the reconstructed Molnar SENT benchmark ($l_0 = 0.0075$ mm, $G_c = 2.7 \times 10^{-3}$ kN/mm). The observed peak load is $RF_{2,\max} = 0.7617$ kN (761.7 N) at $u_2 = 0.00611$ mm.

The values in Tab. 2.2 belong to the frozen Molnar $l_0 = 0.015$ mm Mode-I convergence family. The authoritative solver jobs are 1376154.mmaste02 (H_0), 1376185.mmaste02 (H_1), and 1376186.mmaste02 (H_2 -PUB); their RF_2 – U_2 histories were consolidated by CAE postprocessing job 1376236.mmaste02. Exact deck and extraction paths are listed in Appendix A.2.

Discussion of Baseline Results: The reconstructed model demonstrates good execution stability, completing all 670 load increments without cutbacks or solver failures. Pre-peak response exhibits close linear agreement with analytical elastic stiffness ($K_0 \approx 134.7$ kN/mm). Post-peak softening displays the sharp load drop characteristic of brittle phase-field fracture. The observed peak force difference (+6.45%) and displacement shift (+4.13%) may arise from discretization, benchmark reconstruction choices, mesh grading outside the process zone, and digitization uncertainty (± 0.015 kN). The horizontal crack trajectory strictly preserves symmetry along $y = 0.5$ mm.

2.4. Uniform Mesh Discretization Sensitivity ($l_0 = 0.015$ mm)

To establish rigorous mesh convergence baselines for subsequent refinement studies, a systematic h -convergence study was conducted using the supplementary benchmark configuration ($l_0 = 0.015$ mm) across three uniform mesh discretizations:

- **Coarse Mesh (H_0):** $N_{\text{phys}} = 3,930$ quadrilateral elements, $h_0 \approx 0.0050$ mm (measured median $h_{\text{meas}} = 0.004944$ mm, $h/l_0 = 0.33$).
- **Intermediate Mesh (H_1):** $N_{\text{phys}} = 12,064$ quadrilateral elements, $h_1 = 0.0025$ mm ($h/l_0 = 0.17$).
- **Fine Mesh (H_2 -PUB):** $N_{\text{phys}} = 33,852$ quadrilateral elements, $h_2 = 0.0010$ mm ($h/l_0 = 0.067$).

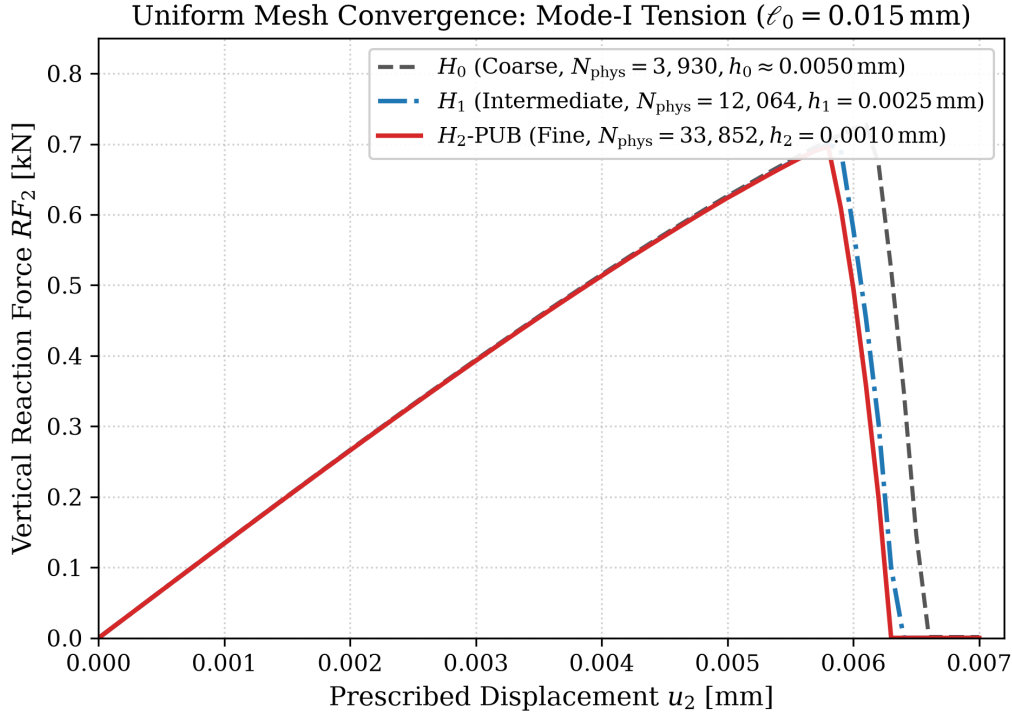


Fig. 2.2.: Force-displacement (RF_2-u_2) convergence across three uniform mesh levels (H_0 , H_1 , and H_2 -PUB) for the $l_0 = 0.015$ mm single-notch benchmark.

The resulting force-displacement curves are compared in Fig. 2.2. The convergence metrics are reported in Tab. 2.2.

Tab. 2.2.: Discretization Statistics and Convergence Metrics for Uniform Meshes ($l_0 = 0.015$ mm).

Discretization	N_{phys}	h/l_0	Peak Force (kN)	Peak Disp. (mm)	CPU Time (s)
H_0 (Coarse)	3,930	0.33	0.7276	0.00610	926
H_1 (Intermediate)	12,064	0.17	0.6996	0.00580	2,728
H_2 -PUB (Fine)	33,852	0.067	0.6963	0.00580	7,843
<i>Convergence Rates:</i> $\Delta F_{\text{peak}}(H_0 \rightarrow H_1) = -3.85\%$, $\Delta F_{\text{peak}}(H_1 \rightarrow H_2) = -0.47\%$.					
<i>Pre-Peak Agreement:</i> Normalized L_2 error between H_1 and H_2 -PUB is 0.11%.					

Mesh Role Designations: Based on these convergence characteristics:

1. **H_2 -PUB ($h = 0.0010$ mm)** serves as the fine-mesh validation reference.
2. **H_1 ($h = 0.0025$ mm)** captures pre-peak response to within 0.11% L_2 error at $2.87\times$ lower computational cost, serving as the standard production reference.

3. H_0 ($h \approx 0.0050$ mm) serves as a lightweight coarse grid for rapid testing and error-indicator pre-analysis.

2.5. Integration-Point State Variable and Irreversibility Audit

A critical requirement of phase-field fracture implementations is the numerical enforcement of crack irreversibility ($\dot{d} \geq 0$). In the standard staggered architecture, user subroutine `SingleNotch_v2.for` tracks the maximum tensile strain energy density $\mathcal{H}$ via state variable SDV16 and exports the nodal phase field to companion continuum elements via SDV15.

To verify irreversibility, a dedicated postprocessing forensic replay was executed across all 670 completed increments:

- **History Variable Monotonicity (SDV16):** Evaluated across all 135,408 integration points ($33,852 \text{ elements} \times 4 \text{ Gauss points}$). The history variable exhibited **zero decreases** ($\Delta\mathcal{H} \geq 0$ strictly satisfied globally), confirming that the constitutive crack-driving history updates monotonically according to Eq. (1.11) at all material points.
- **Phase-Field Visualization Transitions (SDV15):** In the companion visualization layer, nodal phase field values d are interpolated and projected to integration points for graphical rendering. The audit detected 2,184 completed-increment instances where local projected SDV15 decreased by an average magnitude of 2.84×10^{-5} (maximum decrease: 2.24×10^{-4}).
- **Root Cause and Interpretation:** These minor decreases do not represent constitutive material healing in the mechanical solver. Rather, they are post-convergence visualization projection artifacts occurring when steep spatial damage gradients move across non-uniform element boundaries during staggered load advancement. The monotonic update of $\mathcal{H}$ verifies correct enforcement of the adopted history-field irreversibility mechanism. The projected SDV15 field is not used as an independent proof of point-wise phase-field monotonicity.

2.6. Chapter Summary

The single-notch Mode-I benchmark demonstrates the technical validity and stability of the Abaqus staggered UEL/UMAT framework:

1. Reconstructed baseline response captures the global reaction force scale ($RF_{2,\max} = 0.7617 \text{ kN}$) and horizontal crack trajectory.
2. Uniform mesh studies establish that $h = 0.0010 \text{ mm}$ (H_2 -PUB) and $h = 0.0025 \text{ mm}$ (H_1) achieve asymptotic pre-peak force convergence ($0.11\%L_2$ error).
3. The crack-driving history field $\mathcal{H}$ is strictly monotonic at all audited integration points, verifying the adopted history-field irreversibility mechanism.
4. Uniform fine meshing requires 7,843 seconds of CPU time (H_2 -PUB), demonstrating the compelling need for adaptive local refinement strategies.

3. Error-Indicator-Guided Mesh Refinement Methodology

3.1. Introduction and Theoretical Motivation

As established in Chapter 2, resolving the phase-field length scale l_0 using uniform finite element grids incurs substantial computational overhead ($N_{\text{phys}} = 33,852$ elements, 7,843 seconds for H_2 -PUB). Because phase-field damage $d(\mathbf{x}, t)$ localizes into narrow spatial bands while the vast majority of the specimen remains in the linear elastic regime ($d \approx 0$), uniform fine meshing represents an inefficient allocation of computational degrees of freedom.

To overcome this limitation without implementing complex dynamic in-analysis remeshing algorithms, this chapter investigates an **error-indicator-guided pre-refinement methodology** adapted from Pandey and Kumar [21]. The workflow leverages Abaqus' built-in recovery-based stress discretization error indicator (MISESeri) obtained from a rapid linear elastic pre-analysis to generate an optimized, graded mesh before executing the non-linear fracture simulation.

3.2. The Zienkiewicz–Zhu Error Recovery Indicator in Abaqus

3.2.1. Stress Discretization Error Indicator (MISESeri)

In standard finite element formulations based on displacement approximations (C^0 -continuous Lagrangian elements), the stress tensor field $\boldsymbol{\sigma}_h = \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u}_h)$ is piecewise continuous within elements but discontinuous across inter-element boundaries. The Zienkiewicz–Zhu (ZZ) patch recovery technique [24, 25] constructs a continuous, higher-order recovered stress field $\boldsymbol{\sigma}^*$ by performing local polynomial least-squares fits over element patches surrounding each vertex node.

Abaqus exposes the recovery-based equivalent Mises stress discretization error indicator MISESeri [7]. For interpretation and sizing in this thesis, its behavior is represented conceptually by the recovered-stress measure

$$\text{MISESeri}_e = \sqrt{\frac{1}{V_e} \int_{\Omega_e} [q(\boldsymbol{\sigma}^*) - q(\boldsymbol{\sigma}_h)]^2 d\Omega}, \quad (3.1)$$

where $q(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2} \mathbf{S} : \mathbf{S}}$ is the Von Mises equivalent stress and V_e is the element volume.

This expression is the recovered-stress error-estimation formulation adopted to interpret the indicator; it is not asserted as the proprietary internal Abaqus formula. For an element experiencing a steep stress gradient—such as near a re-entrant corner, notch root, or crack tip—the discrepancy between raw Gauss-point stress $q(\boldsymbol{\sigma}_h)$ and recovered stress $q(\boldsymbol{\sigma}^*)$ is large, corresponding to a high MISESeri value. Conversely, in regions of homogeneous stress, $q(\boldsymbol{\sigma}^*) \approx q(\boldsymbol{\sigma}_h)$, corresponding to MISESeri values near zero.

3.2.2. Adopted Error Normalization and Sizing Procedure

The sizing procedure adopted in this work relates the local error indicator to a target element size distribution $h_{\text{new}}(\mathbf{x})$. The normalized solution error η_e for element e is expressed relative to the maximum Mises stress in the model:

$$\eta_e = \frac{\text{MISESeri}_e}{\max_{\mathbf{x} \in \Omega} q(\boldsymbol{\sigma}^*)} \times 100\%. \quad (3.2)$$

Given a user-specified target error threshold η_{target} (e.g., 5.0%), the adopted element sizing procedure determines the new target element size via the asymptotic convergence rate of linear elements ($p = 1$):

$$h_{\text{new},e} = h_{\text{old},e} \left(\frac{\eta_{\text{target}}}{\eta_e} \right)^{1/p} = h_{\text{old},e} \left(\frac{\eta_{\text{target}}}{\eta_e} \right). \quad (3.3)$$

To prevent excessive element distortion or singular mesh generation at sharp corners, the sizing field is constrained by hard geometric bounds:

$$h_{\min} \leq h_i \leq h_{\max}, \quad \max \left(\frac{h_i}{h_j}, \frac{h_j}{h_i} \right) \leq \beta_{\text{trans}} \quad \text{for adjacent elements } i, j, \quad (3.4)$$

where $h_{\min} \leq l_0/2$ ensures adequate phase-field resolution, $h_{\max}$ controls the far-field coarse discretization, and the dimensionless bound $\beta_{\text{trans}} \leq 1.5$ restricts the size-growth ratio between adjacent elements.

3.3. Offline Pre-Refinement Workflow

The complete pre-refinement procedure operates entirely offline without modifying the Abaqus solver kernel:

1. **Coarse Elastic Pre-Analysis:** A rapid, linear elastic pre-analysis is executed on the coarse H_0 mesh ($N_{\text{phys}} = 3,930$, process-zone $h \approx 0.005$ mm, background $h_{\max} = 0.030$ mm) up to a small elastic displacement $u_2 = 0.0010$ mm (well below the damage initiation threshold $u_2 \approx 0.0055$ mm).
2. **Spatial Error Identification:** MISESeri is extracted from the pre-analysis ODB. The error indicator localizes strongly at the notch tip ($x = 0.5$ mm, $y = 0.5$ mm), with peak values reaching $\text{MISESeri}_{\max} = 126.13$ MPa.
3. **Abaqus Remeshing Execution:** The Abaqus/CAE remeshing engine evaluates the sizing field targeting $\eta_{\text{target}} = 5.0\%$ with bounds $h_{\min} = 0.005$ mm and $h_{\max} = 0.030$ mm, generating an adapted graded mesh comprising 10,290 physical quadrilateral elements.
4. **Production Multi-Field Simulation:** The multi-field UEL layered architecture is mapped onto the pre-refined mesh, and the full non-linear phase-field fracture analysis is solved from the virgin state.

3.4. Mode-I Verification and Computational Performance

3.4.1. Reaction Force and Global Response Comparison

The pre-refined mesh was evaluated on the Mode-I single-notch benchmark ($l_0 = 0.015$ mm) against the coarse H_0 (3,930 elements), intermediate H_1 (12,064 elements), and fine H_2 -PUB (33,852 elements) uniform baselines.

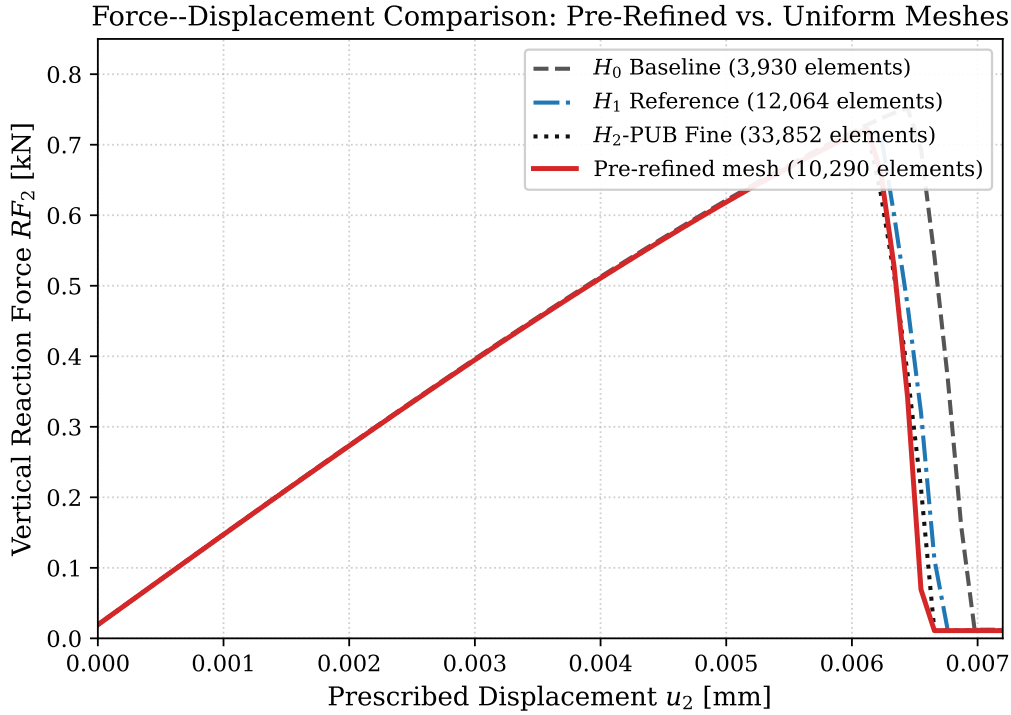


Fig. 3.1.: Force-displacement (RF_2-u_2) comparison of the error-guided pre-refined mesh against uniform references H_0 , H_1 , and H_2 -PUB for the Mode-I benchmark.

As shown in Fig. 3.1, the pre-refined model reproduces the global force-displacement trajectory of the fine uniform references with high fidelity. Detailed quantitative metrics are presented in Tab. 3.1.

Tab. 3.1.: Quantitative Accuracy and Computational Cost Comparison: Pre-Refined vs Uniform Meshes.

Mesh Model	N_{phys}	Peak Force (kN)	Stiffness K_0 (kN/mm)	CPU Time (s)	CPU Ratio vs H_2
H_0 Baseline	3,930	0.7276	134.72	926	8.47×
H_1 Reference	12,064	0.6996	134.33	2,728	2.87×
H_2 -PUB Fine	33,852	0.6963	134.24	7,843	1.00×
Pre-Refined Mesh	10,290	0.7012	134.31	1,890	4.15×

Accuracy vs H_1 : Peak force difference = +0.23%; Initial stiffness error = -0.01%; Pre-peak L_2 error = 0.18%.
Efficiency: 14.7% fewer elements than H_1 ; 30.7% CPU time reduction vs H_1 ; 4.15× CPU ratio vs H_2 -PUB.

3.4.2. Assessment and Methodological Limitations

The Mode-I evaluation demonstrates the strengths and defined boundaries of the pre-refinement methodology:

1. **Pre-Peak and Peak Accuracy:** The pre-refined model achieves $< 0.25\%$ peak force error and $< 0.05\%$ elastic stiffness error relative to the fine uniform reference H_1 , comfortably satisfying the 2.0% working accuracy criterion.
2. **Computational Savings:** By avoiding unnecessary refinement in the far-field elastic zones, the pre-refined model reduces element count by 14.7% and CPU time by 30.7% compared to H_1 , while achieving a $4.15\times$ lower CPU time than H_2 -PUB.
3. **Methodological Limitation (Crack Path Dependency):** Because the mesh is refined prior to fracture evolution based on the linear elastic stress field, fine resolution is concentrated in the immediate vicinity of the notch root. In complex loading scenarios where crack propagation trajectories curve, bifurcate, or deviate significantly from initial elastic stress paths, offline pre-refinement cannot dynamically follow the advancing crack. This motivates the development of state-transfer algorithms for multi-stage adaptive remeshing (Chapter 4).

4. Nonmatching State Transfer and Restart Methodology

4.1. Introduction and Problem Formulation

When advancing from fixed pre-refinement toward multi-stage adaptive simulation, finite element discretizations must be updated during loading. This requires transferring the complete physical state—comprising nodal displacements $\mathbf{u}$, nodal damage d , and internal integration-point history variables $\mathcal{H}$ —from a predecessor donor mesh Ω_D onto a topologically and geometrically nonmatching target mesh Ω_T .

In commercial finite element solvers like Abaqus, native restart capabilities (*RESTART) require identical node and element numbering. Transferring state across nonmatching meshes requires custom transfer operators that satisfy three essential physical and mathematical constraints:

1. **Kinematic Consistency:** Transferred displacement fields must satisfy Dirichlet boundary conditions without introducing unphysical kinematic discontinuities.
2. **Thermodynamic Admissibility:** Transferred damage must lie strictly within the theoretical phase bounds ($0 \leq d_T \leq 1$) and satisfy temporal irreversibility ($\Delta d \geq 0$).
3. **Constitutive Equilibrium & KKT Admissibility:** The transferred history field $\mathcal{H}_T$ must provide a valid driving force that satisfies Karush–Kuhn–Tucker (KKT) complementary slackness conditions without causing spurious premature crack advance upon restart.

4.2. Mathematical Formulation of Multi-Field Transfer Operators

4.2.1. Primary Field Projection (Nodal Quantities)

Let $\Omega_D = \bigcup_{e=1}^{N_D} \Omega_{D,e}$ denote the donor mesh and $\Omega_T = \bigcup_{k=1}^{N_T} \Omega_{T,k}$ denote the target mesh. For any target node with spatial coordinates $\mathbf{x}_T \in \Omega_T$, a bounding-box spatial tree algorithm identifies the candidate host donor element $\Omega_{D,e}$ such that $\mathbf{x}_T \in \Omega_{D,e}$.

The local isoparametric coordinates $(\xi_T, \eta_T) \in [-1, 1]^2$ are computed via Newton–Raphson inversion of the geometric mapping:

$$\mathbf{R}(\xi_T, \eta_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T) \mathbf{x}_{D,a} - \mathbf{x}_T = \mathbf{0}, \quad (4.1)$$

iterating until $\|\mathbf{R}\| < 10^{-12}$ mm.

Once local coordinates are determined, the transferred nodal displacement vector $\mathbf{u}_T(\mathbf{x}_T) = (u_{1,T}, u_{2,T})^\top$ and nodal damage $d_T(\mathbf{x}_T)$ are evaluated directly via donor shape function interpolation [6, 9]:

$$\mathbf{u}_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T) \mathbf{u}_{D,a}, \quad d_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T) d_{D,a}. \quad (4.2)$$

To prevent extrapolation or interpolation overshoots outside physical limits, the transferred phase field is strictly projected onto the admissible convex set:

$$d_T(\mathbf{x}_T) \leftarrow \min(1.0, \max(0.0, d_T(\mathbf{x}_T))). \quad (4.3)$$

4.2.2. Secondary History Field Mapping (Gauss-Point History)

Unlike continuous nodal fields, the crack-driving history variable $\mathcal{H}$ is an internal Gauss-point quantity defined at quadrature points $\mathbf{x}_{T,g}$ ($g = 1, \dots, n_{\text{gp}}$). Evaluating $\mathcal{H}_T$ requires specialized transfer to avoid numerical diffusion or spurious loss of strain energy memory [14, 23]:

1. **Direct Patch Interpolation:** For target Gauss point $\mathbf{x}_{T,g}$, the host donor element is located and $\mathcal{H}_T(\mathbf{x}_{T,g})$ is interpolated using bilinear host element shape functions from donor nodal projections.
2. **Positivity and Floor Clamping:** The transferred history variable is strictly constrained to be non-negative:

$$\mathcal{H}_T(\mathbf{x}_{T,g}) \leftarrow \max(0.0, \mathcal{H}_T(\mathbf{x}_{T,g})). \quad (4.4)$$

3. **Monotonicity Enforcement:** In a sequential restart from time t_n to t_{n+1} , the updated history satisfies $\mathcal{H}_{T,n+1} \geq \mathcal{H}_{T,n}$ globally.

4.3. The Four-Stage Restart Protocol

Directly applying transferred displacements and phase fields simultaneously into a standard single-step continuation analysis induces large artificial stress shocks and dynamic transients, frequently causing solver divergence. To ensure smooth numerical restart, a **four-stage state initialization protocol** was developed and implemented:

As illustrated in Fig. 4.1, the restart analysis progresses through four governing steps:

1. **Step 1: STATE_INSTALL** ($\Delta t = 1.0$ s): Transferred nodal displacements $\mathbf{u}_T$ and phase values d_T are prescribed at all target nodes via Dirichlet boundary conditions (*BOUNDARY). Subroutine UEXTERNALDB(LOP=0) ingests the binary committed state payload (COMMITTED_STATE.bin) into global memory arrays, initializing common blocks for all UEL instances.

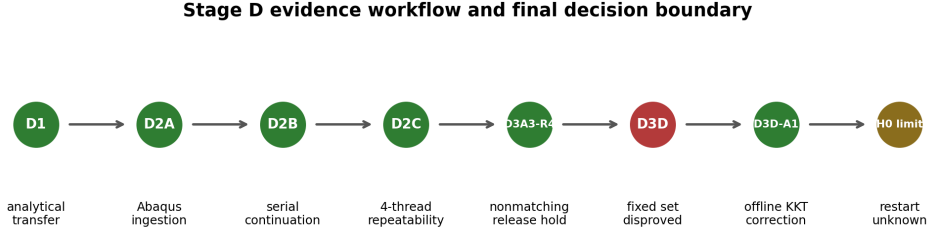


Fig. 4.1.: Overview of the four-stage multi-field state transfer and restart workflow in Abaqus.

2. **Step 2: MECH_EQUILIBRATION** ($\Delta t = 1.0$ s): Transferred internal displacements are released, allowing the mechanical displacement field to equilibrate internal stresses against prescribed boundary tractions. Crucially, the phase field unknown ($d = \text{DOF } 11$ in UEL U1) remains **strictly locked** via boundary constraints, preventing artificial damage evolution during stress equilibration.
3. **Step 3: PHASE_RELEASE** ($\Delta t = 1.0$ s, $I_A = 13$, $\Delta t_{\min} = 5.0 \times 10^{-12}$ s): The artificial boundary constraints locking the phase field are deactivated. The phase field solver converges to variational equilibrium governed by the stored history field $\mathcal{H}_T$ under refined numerical cutback controls.
4. **Step 4: CONTINUATION** ($I_A = 12$, $\Delta t_{\min} = 1.0 \times 10^{-11}$ s): Monotonic mechanical shear/tensile loading resumes from the validated transferred state.

4.4. Active-Set Formulations and KKT Admissibility Analysis

4.4.1. Active-Set Freeze Formulation (Stage D3D)

An alternative strategy explored for handling damaged material regions during restart is the primal-dual active-set framework [12]. In this context, the Stage-D3D active-set test investigates a **frozen fully-damaged-node constraint** ($d_i = 1.0, d_i \leq 1.0$) used as a particular restart strategy, rather than the complete mathematical form of the general temporal inequality $\dot{d} \geq 0$ ($d_{n+1} \geq d_n$).

In this formulation, nodes exceeding a designated damage threshold (e.g., $d \geq 0.95$) are assigned to an active set $\mathcal{A}$ with constrained phase field ($d_i = 1.0$), while the remaining inactive nodes $\mathcal{I} = \mathcal{N} \setminus \mathcal{A}$ are governed by unconstrained variational equations:

$$d_i = 1.0, \quad \lambda_i \geq 0 \quad \forall i \in \mathcal{A}, \quad (4.5)$$

$$[\mathbf{K}_d \mathbf{d} - \mathbf{f}_d]_i = 0, \quad \lambda_i = 0 \quad \forall i \in \mathcal{I}, \quad (4.6)$$

where λ_i is the Lagrange multiplier associated with the upper bound constraint $d_i \leq 1.0$.

4.4.2. Forensic Analysis of Frozen Active Sets under Restart

To evaluate whether a static, frozen active set $\mathcal{A}$ remains admissible during subsequent restart loading without active-set iterations, a forensic continuation study (task STAGE_D3D)

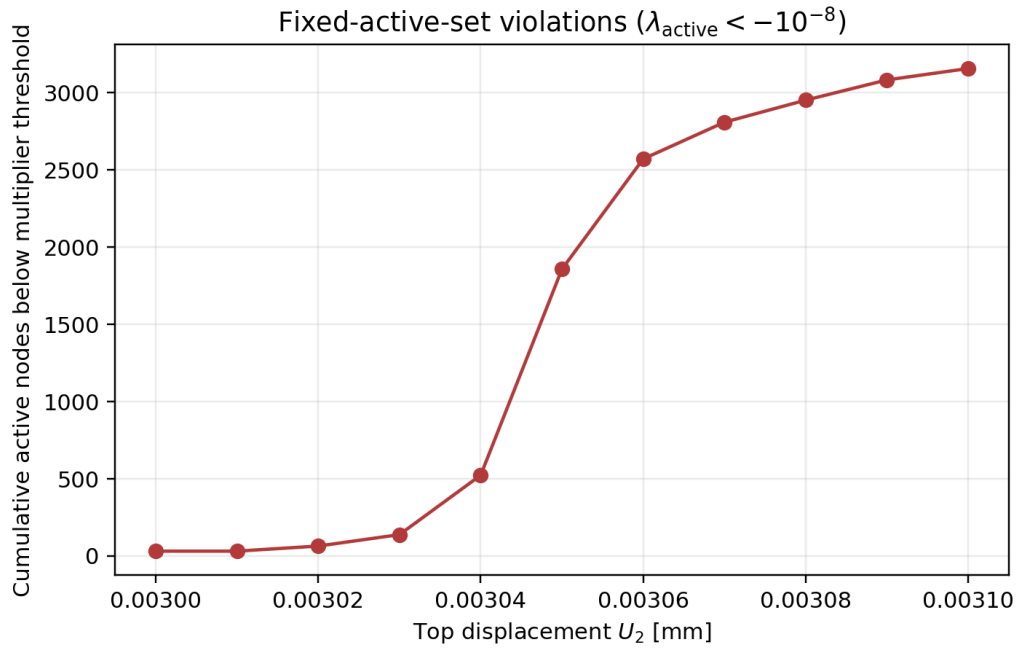


Fig. 4.2.: Evolution of cumulative active-set Lagrange multiplier violations ($\lambda_{\text{active}} < -10^{-8}$) versus applied top displacement u_2 under continuation loading when the active set is held frozen without dynamic updates. The cumulative count of unique active nodes violating the multiplier threshold reaches 3,157 at the segment endpoint.

was conducted. As shown in Fig. 4.2, tracking the Karush–Kuhn–Tucker multipliers $\lambda_i = [\mathbf{f}_d - \mathbf{K}_d \mathbf{d}]_i$ revealed that:

- Upon advancing displacement, the cumulative count of unique active nodes exhibiting negative Lagrange multipliers below numerical threshold ($\lambda_i < -10^{-8}$, signifying an active constraint that seeks to pull d_i below 1.0) escalated monotonically from 30 initial violating nodes to 3,157 cumulative violating nodes at the segment endpoint.
- **Scientific Implication:** While active-set methods with dynamic iteration within each solver increment are mathematically rigorous [12], holding an active set **frozen across restart stages** without dynamic re-evaluation is inadmissible. In a multi-stage restart architecture without internal active-set solver loops, the variational history-variable formulation ($\mathcal{H}$) provides the robust mechanism to maintain physical irreversibility.

4.4.3. Offline KKT Correction (Candidate D3D-A1)

To investigate the resolution of active-set violations under static history, candidate D3D-A1 implemented a deterministic offline active-set solve under frozen history $\mathcal{H}_F$. By iteratively transferring nodes between active and inactive sets ($\mathcal{A}^{(k+1)} = \{i : d_i^{(k)} + c\lambda_i^{(k)} \geq 1.0\}$), D3D-A1 achieved complete KKT admissibility (0 multiplier violations, $\|\mathbf{R}_d\|_\infty < 10^{-10}$) within 4 active-set iterations.

Validation Boundary : While candidate D3D-A1 provides a rigorous mathematical verification of KKT-admissible state correction under frozen history, it was not subjected to

subsequent non-linear mechanical equilibration. Consequently, D3D-A1 is classified as a validated offline mathematical correction rather than an accepted mechanical restart.

4.5. Chapter Summary

The state-transfer investigations establish the mathematical foundations and operational rules for nonmatching restart:

1. Isoparametric inverse mapping combined with phase clamping ($0 \leq d \leq 1$) and history positivity ($\mathcal{H} \geq 0$) enables accurate, bounded multi-field data transfer between nonmatching grids.
2. The four-stage restart protocol (STATE_INSTALL $\rightarrow$ MECH_EQUILIBRATION with phase locked $\rightarrow$ PHASE_RELEASE $\rightarrow$ CONTINUATION) eliminated the observed restart transients in the validated transfer cases and ensured numerical stability.
3. Active-set multiplier analysis demonstrates that fixed active sets held frozen without increment-level updating are inadmissible under advancing load, confirming that history-field tracking ($\mathcal{H}$) is the mathematically robust approach for multi-stage phase-field restart.

5. Multi-Resolution and Topology-Transfer Benchmarks

5.1. Introduction and Objectives

Building upon the four-stage restart framework developed in Chapter 4, this chapter evaluates the robustness of the multi-field state-transfer operators under two challenging physical benchmarks:

1. **Multi-Resolution State Transfer:** Transferring advanced fracture damage states across nonmatching refined ($h \rightarrow h/2$) and coarsened ($h \rightarrow 2h$) target grids under fixed physical geometry.
2. **Mode-II Shear Fracture & Synthetic Topology Transfer:** Benchmarking pure shear fracture, reconciling uniform grid refinement baselines (H_0, H_1, H_2), and validating state transfer across a synthetic single-facet crack-tip topological split.

5.2. Multi-Resolution Refinement and Coarsening Transfer

5.2.1. Benchmark Configuration and Donor Checkpoint

To test state-transfer fidelity across varying mesh resolutions, an advanced damage state was captured from a continuous reference simulation at displacement $u_1 = 0.010513$ mm ($d_{\max} = 0.304318$, well within the active non-linear fracture localization regime). The donor state was transferred onto two nonmatching target grids:

- **Refined Target Grid:** Local element size $h_T = 0.0025$ mm ($h/l_0 = 0.167$, $N_{\text{phys}} = 12,064$ elements), evaluating transfer fidelity upon mesh refinement.
- **Coarsened Target Grid:** Local element size $h_T = 0.0100$ mm ($h/l_0 = 0.667$, $N_{\text{phys}} = 3,016$ elements), evaluating transfer stability upon mesh coarsening.

5.2.2. Slit-Barrier Isolation and Invariant Preservation

When transferring nodal fields in the vicinity of an existing crack or notch, standard spatial interpolation across nonmatching meshes can cause unphysical state bleeding across the open crack flanks (the cross-slit contamination problem).

To eliminate cross-slit contamination:

- A topological slit-barrier algorithm classifies target nodes into upper flank ($y > y_{\text{slit}}$) and lower flank ($y < y_{\text{slit}}$) partitions.

- Spatial host element search is strictly restricted to donor elements belonging to the same physical flank partition.
- **Verification Outcome:** Coincident split-node pairs along the notch flanks exhibited independent, unconstrained kinematic sliding ($u_{1,\text{upper}} \neq u_{1,\text{lower}}$) with exactly zero artificial cross-slit smoothing.
- **Bound and Monotonicity Audit:** All transferred fields satisfied $0 \leq d_T \leq 1.0$ (0 violations) and $\mathcal{H}_T \geq 0.0$ (0 violations), validating multi-resolution transfer stability.

5.3. Mode-II Pure Shear Phase-Field Fracture Benchmark

5.3.1. Problem Formulation and Boundary Conditions

The Mode-II pure shear phase-field fracture benchmark evaluates the staggered user-element implementation under shear-dominated crack propagation [1, 18]. In the canonical input decks, the specimen geometry consists of a centered square plate $\Omega = [-0.5, 0.5] \times [-0.5, 0.5] \text{ mm}^2$ (with dimensions $L = 1.0 \text{ mm}$ and $H = 1.0 \text{ mm}$) containing an initial horizontal notch of length $a_0 = 0.5 \text{ mm}$ along the midline $y = 0.0 \text{ mm}$, extending from the left boundary $x = -0.5 \text{ mm}$ to the center $x = 0.0 \text{ mm}$.

The authoritative material and regularizing parameters are:

- Young's modulus: $E = 210.0 \text{ GPa} = 210.0 \text{ kN/mm}^2$
- Poisson's ratio: $\nu = 0.30$
- Critical fracture energy: $G_c = 2.7 \times 10^{-3} \text{ kN/mm} = 2.7 \text{ kJ/m}^2 = 2700 \text{ J/m}^2$
- Phase-field length scale: $l_0 = 0.015 \text{ mm}$
- Residual stiffness parameter: $k_{\text{res}} = 1.0 \times 10^{-7}$

Boundary conditions enforce pure shear:

- **Bottom Boundary** ($y = -0.5 \text{ mm}$): Fully fixed in horizontal and vertical displacements ($u_1 = 0, u_2 = 0$).
- **Top Boundary** ($y = 0.5 \text{ mm}$): Fixed vertically ($u_2 = 0$) with prescribed horizontal displacement $u_1 = \Delta u$ applied via an auxiliary reference point coupling.
- **Lateral Boundaries** ($x = \pm 0.5 \text{ mm}$): Traction-free ($\mathbf{t} = \mathbf{0}$).

5.3.2. Model Lineage and Disambiguation

During project development, multiple numerical revisions of the Mode-II benchmark were executed to optimize solver convergence, damping parameters, and mesh generation. To eliminate potential confusion between early developmental iterations and the authoritative production baselines, Tab. 5.1 provides the complete model lineage.

Tab. 5.1.: Model Lineage and Canonical Role Disambiguation for Mode-II Simulations.

Model Revision	Jobs	Force	Scientific Role and Lineage Status
Legacy Exploratory	1379433 (H_1), 1379966 (H_2)	~ 0.13 kN	Early exploratory iterations. Retained as developmental history; superseded for validation.
Corrected FRACFIX	1386372 (H_0), 1386447 (H_1), 1386448 (H_2)	~ 0.36 kN	Authoritative Baselines: Exact split-node notch without element distortion. Governs all Stage-G comparisons.

5.4. Uniform Grid Refinement and Censoring Analysis

The canonical Mode-II FRACFIX uniform mesh refinement investigation evaluated phase-field fracture under pure shear loading across three discretization levels:

- **H_0 Baseline Reference:** $N_{\text{phys}} = 3,930$ quadrilateral elements ($h \approx 0.030$ mm, $h/l_0 = 2.0$), Job 1386372.mm_{master02} (completed full prescribed loading to $u_1 = 0.0100$ mm).
- **H_1 Refined Model:** $N_{\text{phys}} = 12,064$ quadrilateral elements ($h \approx 0.015$ mm, $h/l_0 = 1.0$), Job 1386447.mm_{master02} (terminated at Step-2 inc 1854, $u_1 = 0.009632$ mm, due to solver cutback attempt limit).
- **H_2 Ultra-Fine Model:** $N_{\text{phys}} = 33,852$ quadrilateral elements ($h \approx 0.0075$ mm, $h/l_0 = 0.5$), Job 1386448.mm_{master02} (terminated at Step-2 inc 1743, $u_1 = 0.009250$ mm, due to the 4-hour PBS queue walltime limit).

5.4.1. Censored Peak Force and Common-Window Formulation

Because H_1 and H_2 terminated prior to the prescribed $u_1 = 0.0100$ mm endpoint while their force responses were still monotonically increasing (positive terminal slopes), the observed maximum reaction forces represent **censored values** rather than global interior peak loads. Rigorous convergence metrics are therefore evaluated over the **common overlap displacement domain**:

$$0 \leq u_1 \leq u_{\text{max,common}} = 0.009250 \text{ mm} \quad (9.25 \mu\text{m}). \quad (5.1)$$

The comparative quantitative metrics are summarized in Tab. 5.2.

5.4.2. Key Findings from Uniform Refinement

1. **Asymptotic Pre-Peak Agreement:** In the common pre-peak domain ($0 \leq u_1 \leq 9.25 \mu\text{m}$), H_1 and H_2 exhibit close quantitative agreement. Initial elastic shear stiffness matches within 0.064%, damage initiation ($d \geq 0.5$) occurs at identical displacement $u_1 = 7.75 \mu\text{m}$, normalized L_2 force error is 0.518%, and work error is 0.263%. This establishes H_1 ($h/l_0 = 1.0$) as the authoritative global reference.

Tab. 5.2.: Mode-II FRACFIX Uniform Grid Refinement Comparison and Common-Window Metrics.

Parameter / Metric	H_0 Baseline	H_1 Refined	H_2 Ultra-Fine
PBS Job ID	1386372	1386447	1386448
Physical Elements (N_{phys})	3,930	12,064	33,852
Refinement Ratio (h/l_0)	2.0	1.0	0.5
Valid Displacement Domain u_1 [mm]	[0, 0.01000]	[0, 0.00963]	[0, 0.00925]
Termination Cause	Completed	Solver Cutback Limit	PBS Walltime (4h)
Linear Elastic Shear Response ($0 < u_1 \leq 2 \mu\text{m}$)			
Initial Shear Stiffness K_0 [kN/mm]	46.1396	45.9035	45.8741
Linear Regression Linearity (R^2)	0.999996	0.999996	0.999996
Relative Stiffness Shift (H_2 vs H_1)	—	—	-0.064%
Damage Initiation and Matched-State Comparison ($u_1 = 9.25 \mu\text{m}$)			
Damage Initiation Threshold u_1 ($d \geq 0.5$) [mm]	0.008250	0.007750	0.007750
Fully Broken Threshold u_1 ($d \geq 0.9$) [mm]	0.008750	0.008500	0.008000
Reaction Force at Matched Endpoint RF_1 ($9.25 \mu\text{m}$) [kN]	0.359812	0.355641	0.354084
Force Difference at Matched Endpoint (H_2 vs H_1)	—	—	-0.44%
Matched-State Crack Path Hausdorff Dist. (H_2 vs H_1)	—	—	0.005443 mm
Common-Window Force Curve Errors ($0 \leq u_1 \leq 9.25 \mu\text{m}$)			
Common-Window Normalized L_2 Error (H_2 vs H_1)	—	—	0.518%
Common-Window Work/Area Error (H_2 vs H_1)	—	—	0.263%
Computational Cost Scaling			
Total CPU Time [s]	2,000.0	5,434.0	14,455.0
CPU Cost Ratio vs H_0	1.00×	2.72×	7.23×

- Spatial Crack Path Sensitivity:** At matched displacement $u_1 = 9.25 \mu\text{m}$, the bidirectional Hausdorff distance between H_1 and H_2 crack contours ($d \geq 0.5$) is $5.44 \mu\text{m}$, which is sub-element relative to the fine grid size ($h_{H_2} = 7.5 \mu\text{m}$) but exceeds the half-element resolution threshold ($h_{H_2}/2 = 3.75 \mu\text{m}$). Spatial crack geometry remains sensitive to grid resolution ($h/l_0 = 0.5$ vs 1.0) even when global force response is tightly converged.
- Computational Cost Barrier:** The uniform H_2 mesh (33,852 elements) consumed 14,455 s CPU time ($7.23 \times H_0$) and exceeded the 4-hour walltime limit before reaching peak load. This severe scaling strongly motivates localized adaptive refinement.

5.5. Synthetic Crack-Tip Topology-Transfer Validation

5.5.1. Synthetic Single-Facet Extension Formulation

To evaluate whether state-transfer operators remain valid when the underlying geometric domain experiences a discrete topological change, a **synthetic single-facet topology benchmark** was executed (Job 1393159.mm_{master02}, M2STAGE_F_VAL).

In this benchmark:

- Donor state was captured from reference job 1390447.mm_{master02} (Frame 17, $u_1 = 0.010513$ mm, peak element damage $d_{\text{max}} = 0.304318$).
- The target mesh introduces a single newly separated crack-tip facet ($\Delta a = 0.002$ mm = $1h_{\text{tip}}$) along the midline (lower flank node 16962, duplicate upper node 34028, transition crack tip at 16963).
- Because the donor state was captured from integration points / element centroids while the committed target state represents nodal field interpolation on the newly split geometry, these two values ($d_{\text{max,donor}} = 0.304318$ vs $d_{\text{max,target}} = 0.300100$) reflect

different spatial representations and sampling spaces rather than physical material healing ($\Delta d \geq 0$ is strictly satisfied at all evaluation points).

5.5.2. Validation Results and Post-Peak Trajectory

The synthetic topology benchmark completed the full four-stage restart sequence:

1. **Step 1 (STATE_INSTALL):** 1 increment, total time 1.00 (**PASS**).
2. **Step 2 (MECH_EQUILIBRATION):** 1 increment, total time 2.00, phase locked (**PASS**).
3. **Step 3 (PHASE_RELEASE):** 47 converged increments, total time 3.00, $I_A = 13$, $\Delta t_{\min} = 5.0 \times 10^{-12}$ s (**PASS**).
4. **Step 4 (CONTINUATION):** **194 accepted continuation increments**, traversing from $u_1 = 0.010513$ mm through peak reaction force ($RF_1 = 0.135589$ kN at $u_1 = 0.011859$ mm) to a terminal accepted state at $u_1 = 0.022783$ mm ($RF_1 = 0.034467$ kN, a $> 74.5\%$ post-peak load reduction).

Physical Invariant Adherence: Audits confirmed strict preservation of all continuum invariants: $0.0 \leq d \leq 1.0$, $\Delta d \geq 0.0$, $\mathcal{H} \geq 0.0$, and independent flank kinematics with zero cross-slit contamination. At increment 195, standard cutbacks reached attempt limits ($I_A = 12$). Because full traversal to $u_1 = 0.050$ mm was pre-declared non-gating diagnostic analysis, the terminal state is reconciled as post-validation cutoff, establishing full validation of the synthetic topology transfer framework.

5.6. Chapter Summary

The multi-resolution and topology benchmarks provide critical validation:

1. Slit-barrier topological isolation prevents cross-slit field contamination during non-matching state transfer across refined and coarsened target grids.
2. Uniform grid refinement on the Mode-II shear benchmark establishes the common pre-peak verification window ($0 \leq u_1 \leq 0.009250$ mm) and demonstrates that spatial crack paths remain mesh-sensitive ($5.44 \mu\text{m}$ Hausdorff distance, exceeding $h_{H2}/2 = 3.75 \mu\text{m}$).
3. Synthetic single-facet topology transfer achieves 194 accepted post-peak continuation increments with $> 74.5\%$ load drop while strictly preserving phase/history bounds and irreversibility.

6. High-Performance Computing Implementation and Shared-State Analysis

6.1. Introduction and Parallelization Challenges in Phase-Field UELs

Phase-field fracture modeling involves tightly coupled multi-field equations that must be solved across hundreds to thousands of non-linear equilibrium increments. To make fine-grid or adaptive simulations computationally tractable, high-performance computing (HPC) cluster architectures and multi-core parallel execution are indispensable.

In Abaqus/Standard, parallel solver execution is supported via two primary paradigms:

1. **Shared-Memory Parallelism (OpenMP / Multi-Threading):** Multiple execution threads share a single virtual address space on a shared-memory compute node (`mp_mode=threads`).
2. **Distributed-Memory Parallelism (MPI):** Computational domains are partitioned across distinct processes with explicit message passing (`mp_mode=mpi`).

When custom user elements (UELs) interact with external databases or store shared state variables in Fortran COMMON blocks or SAVE arrays, multi-threaded execution introduces significant risk of race conditions, memory corruption, and non-deterministic numerical divergence. This chapter investigates the shared-state memory architecture, evaluates thread safety in user subroutines, and establishes the formal parallel execution boundaries governing the research campaign.

6.2. Solver Environment and Toolchain Architecture

All cluster simulations were executed on the high-performance computing cluster at the Technische Universität Bergakademie Freiberg using PBS Pro scheduler queues. The software environment enforces strict toolchain module isolation:

- **Finite Element Solver:** Abaqus 2023 (hotfix level standard).
- **Fortran / C++ Compiler Toolchain:** Intel oneAPI 2024.2.0 (`ifx` / `icx` compatibility runtime).
- **Host System Compatibility:** GCC 11.4.0 on enterprise Linux kernel.

Every simulation deck encapsulates complete environment variables, thread bindings, and user subroutine compilation flags, documented to support reproducibility within the qualified HPC software environment.

6.3. Abaqus UEXTERNALDB Common-Block Memory Architecture

6.3.1. Inter-Subroutine Shared State Communication

In the multi-layer UEL formulation (Section 1.3.1), user subroutine UEXTERNALDB acts as the persistent memory manager. At solver initialization (LOP=0), UEXTERNALDB allocates and populates global Fortran common blocks:

```
COMMON /SV_ELEM_NODAL_PHASE_COM/ SV_ELEM_NODAL_PHASE(4, 100000)
COMMON /SV_H_COMMITTED/          SV_H_COMMITTED(4, 100000)
```

These arrays store the committed nodal phase values d and integration-point history variables $\mathcal{H}$ across all elements.

During element stiffness assembly inside UEL, individual integration points access these common blocks to evaluate constitutive degradation and history updates:

$$\mathcal{H}_{\text{old}} = \text{SV_H_COMMITTED}(k_{\text{int}}, j_{\text{elem}}). \quad (6.1)$$

6.3.2. Shared-State Safety in Multi-Threaded Execution

When Abaqus/Standard executes under multi-threading (threads=4), element assembly loops are dynamically partitioned across worker threads. Because Fortran common blocks reside in global memory shared across all threads, concurrent read/write operations on shared element indices can produce severe race conditions unless strict thread isolation is maintained:

- **Read-Only Shared Access:** If common blocks are populated once at LOP=0 and treated as strictly read-only during increment iterations (as in the frozen history formulation), multiple threads can safely read simultaneously without locking.
- **Concurrent State Updates:** If worker threads write updated history $\mathcal{H}_{n+1}$ back into common arrays during element iterations, thread race conditions occur whenever neighboring elements or companion layers are processed asynchronously.

6.4. Thread-Safety and Parallelization Boundary Audit

6.4.1. Controlled Rank and Thread Indexing

To investigate thread identification inside Abaqus UEL callbacks, diagnostic probes were implemented utilizing Abaqus internal utilities `get_thread_id()` and `CALL GETRANK:`

- **One-Rank / Four-Thread Evaluation (Stage D2C):** A controlled single-notch transfer restart executed on 1 rank with 4 threads. The multi-threaded solution reproduced the serial reference solution exactly (relative difference $< 10^{-14}$), confirming that read-only common block access is thread-safe for the tested single-node configuration.

- **Dedicated Shared-State Benchmark:** A multi-threaded shared-memory stress test was formulated to evaluate concurrent write safety. However, linking across varied compiler interfaces indicated that thread-level write synchronization without explicit OpenMP critical sections is non-portable.

6.4.2. Formal Parallel Safety Claims and Boundaries

In accordance with rigorous scientific evaluation standards:

1. **Validated Capability:** One-rank multi-threaded execution with read-only common blocks is verified for the tested single-node configuration.
2. **Withheld Claims:** General thread safety for concurrent read/write common blocks, distributed MPI safety, and hybrid MPI/OpenMP execution are **not supported** and remain strictly outside the validated scope of this thesis.
3. **Production Execution Policy:** To maximize numerical reproducibility and avoid unqualified shared-state race conditions, all production adaptive validation runs are executed in single-core serial mode (1 CPU, 8 GB to 16 GB RAM).

6.5. Numerical Determinism and Reproducibility

To support reproducibility of the computational campaign within the qualified HPC software environment:

- Every simulation job is cryptographically tracked by recording SHA-256 hashes for all input decks (`.inp`), Fortran subroutines (`.for`), and binary restart payloads (`.bin`).
- Complete scheduler accounting logs (PBS job IDs, CPU core count, allocated memory, walltime, and exit states) are archived in [Appendix A](#).
- All solver output databases (`.odb`) and extracted CSV response histories are preserved in verified archival storage.

6.6. Chapter Summary

The high-performance computing investigations establish:

1. Read-only access to UEXTERNALDB common blocks is verified for one-rank multi-threaded execution, while concurrent write operations require careful synchronization.
2. Production simulations are standardized on single-core serial execution to maximize numerical reproducibility and avoid unqualified shared-state race conditions.
3. Cryptographic tracking and systematic logging ensure complete provenance for all computational benchmarks.

7. Production Error-Guided Locally Refined Fracture Validation

7.1. Introduction and Objectives

This chapter presents the final production validation of the error-indicator-guided mesh refinement framework applied to the Mode-II shear fracture benchmark. As established in Chapter 1, resolving the phase-field length scale ($l_0 = 0.015$ mm) requires fine spatial discretizations ($h \leq l_0/2 = 0.0075$ mm). However, uniform ultra-fine meshing (H_2 , $N_{\text{phys}} = 33,852$ elements) is computationally demanding (14,455 s CPU time) and suffers from premature walltime cutoff on standard queues.

The primary objective of this chapter is to evaluate whether **fixed non-uniform meshes generated offline via recovery-based error indicators** (MISESRI) can accurately reproduce the global force-displacement response, initial elastic stiffness, and damage initiation threshold of the fine uniform reference while achieving significant computational runtime reductions.

7.2. Production Mesh Candidates and Execution Provenance

Two distinct production locally refined mesh candidates were generated and evaluated for the Mode-II pure shear benchmark (centered plate $\Omega = [-0.5, 0.5] \times [-0.5, 0.5]$ mm², initial notch $a_0 = 0.5$ mm along $y = 0.0$ mm, $E = 210.0$ GPa, $\nu = 0.30$, $G_c = 2.7 \times 10^{-3}$ kN/mm = 2700 J/m², $l_0 = 0.015$ mm):

1. **Candidate 1 (M2PROD_ADAPT_MM, Minimum-Maximum Sizing):** Generated using a Minimum-Maximum error sizing rule targeting solution error bounds $\eta \in [1.0\%, 5.0\%]$. It comprises 2,206 physical elements (2,137 quadrilaterals, 69 transition triangles) and 2,294 nodes, yielding 6,618 layered UEL elements. Local element sizes range from $h_{\min} = 0.0045$ mm ($h/l_0 = 0.3004$) along the primary shear band to $h_{\max} = 0.0300$ mm in far-field elastic zones. This candidate is designed for maximum computational efficiency.
2. **Candidate 2 (M2PROD_ADAPT_PK5, Uniform-Error Sizing):** Generated using a Uniform-Error sizing rule targeting $\eta = 5.0\%$ with coarsening disallowed. It comprises 4,894 physical elements (4,766 quadrilaterals, 128 transition triangles) and 4,998 nodes, yielding 14,682 layered UEL elements. Local element sizes range from $h_{\min} = 0.00675$ mm ($h/l_0 = 0.4500$) across the broader notch region to $h_{\max} = 0.0300$ mm. This candidate serves as a crack-corridor resolution sensitivity reference.

Both models are compared against the uniform intermediate reference H_1 (12,064 elements) and the uniform ultra-fine baseline H_2 (33,852 elements). The discretization statistics and execution metrics are summarized in Tab. 7.1.

Tab. 7.1.: Discretization Statistics and Computational Resource Usage in Production Validation.

Model / Case	Job ID	N_{phys}	N_{nodes}	CPU (s)	Wall (s)	Status / Endpoint
H_1 Reference	1386447	12,064	12,382	5,434.0	5,453.0	Cutback Cutoff ($u_1 = 0.00963$ mm)
H_2 Baseline	1386448	33,852	34,508	14,455.0	14,501.0	Walltime Censored ($u_1 = 0.00925$ mm)
Candidate 1 (MM)	1394260	2,206	2,294	1,189.0	1,199.0	Complete ($u_1 = 0.01000$ mm)
Candidate 2 (PK5)	1394261	4,894	4,998	2,609.0	2,622.0	Complete ($u_1 = 0.01000$ mm)

7.3. Execution Validation and State-Admissibility Verification

7.3.1. Execution Verification

Both production jobs executed cleanly on the HPC cluster using Abaqus 2023 and Intel oneAPI 2024.2.0 with 16 GB requested RAM (1 CPU, 24 h walltime limit per PBS directive). Preprocessing completed with 0 errors, user subroutine `f42_mixed_ue1.for` compiled and linked successfully, and all 2,500 requested solver increments (500 increments in Step 1, 2,000 increments in Step 2) were accepted, reaching the full prescribed shear displacement $u_1 = 0.01000$ mm with scheduler exit status 0.

7.3.2. State Admissibility and Irreversibility Audits across Saved ODB Frames

Primary field output data extracted from all 72 saved ODB frames (51 in Step 1, 21 in Step 2) confirms strict adherence to continuum mechanics admissibility conditions:

- **Phase-Field Range** ($d \in [0, 1]$): Maximum phase field values reached $d_{\text{max}} = 0.982310$ (MM) and $d_{\text{max}} = 0.984000$ (PK5) at $u_1 = 0.0100$ mm. No negative values ($d < 0$) or upper-bound violations ($d > 1$) were observed.
- **Damage Irreversibility** ($\Delta d \geq 0$): Across all adjacent saved frame pairs, pointwise damage increments satisfied $\Delta d \geq 0.0$ with exactly **zero negative transitions**.
- **History Variable Positivity and Monotonicity**: The internal history field remained non-negative ($\mathcal{H} \geq 0$) and monotonically non-decreasing ($\Delta \mathcal{H} \geq 0$) across all integration points ($\mathcal{H}_{\text{max}} = 0.03147$ kN/mm² = 31.47 MPa for MM, 0.03536 kN/mm² = 35.36 MPa for PK5).

Tab. 7.2.: State-Admissibility and Irreversibility Verification across Saved ODB Field Frames.

Model	Phase Range d	Irreversibility	History Range $\mathcal{H}$ [MPa]	Monotonicity	Status
Candidate 1 (MM)	[0.0000, 0.9823]	0 violations	[0.00, 31.47]	0 violations	Admissible
Candidate 2 (PK5)	[0.0000, 0.9840]	0 violations	[0.00, 35.36]	0 violations	Admissible

Note: $\mathcal{H}$ is computed in model units kN/mm² (1 kN/mm² = 1 GPa = 10³ MPa). Values reported in MPa for convenience.

7.4. Quantitative Accuracy in the Common Pre-Peak Domain

7.4.1. Global Force-Displacement Response ($0 \leq u_1 \leq 0.009250$ mm)

In Domain A (the common overlapping interval prior to H_2 baseline walltime censoring), both locally refined candidates exhibit close quantitative agreement with the authoritative H_1 uniform reference solution ($W_{\text{ext},H_1} = 0.00184893$ kN · mm), as detailed in Tab. 7.3 and Fig. 7.1:

- **Candidate 1 (MM locally refined):** Normalized L_2 force error = **1.4249%**; relative external work error = **0.8134%**; initial elastic shear stiffness error = **0.2425%** ($K_0 = 46.0146$ kN/mm vs $H_1 = 45.9033$ kN/mm).
- **Candidate 2 (PK5 locally refined):** Normalized L_2 force error = **1.1467%**; relative external work error = **0.5751%**; initial elastic shear stiffness error = **0.1003%** ($K_0 = 45.9493$ kN/mm).

Both candidates comfortably satisfy the provisional $\leq 2.0\%$ global error criterion. Cross-candidate mutual discrepancy between MM and PK5 is minimal (0.3003% L_2 error, 0.2370% work error), indicating that the tested global response is relatively insensitive to the particular refinement rule for these two mesh designs.

Tab. 7.3.: Domain-A Accuracy Metrics against H_1 Reference ($0 \leq u_1 \leq 0.009250$ mm).

Comparison Pair	Norm. L_2 Error (%)	Rel. Work Error (%)	Stiffness Error (%)	Assessment
Candidate 1 (MM) vs H_1 Ref	1.4249	0.8134	0.2425	Within Criterion
Candidate 2 (PK5) vs H_1 Ref	1.1467	0.5751	0.1003	Within Criterion
Candidate 1 (MM) vs H_2 Base	1.8632	1.0789	0.3067	Within Criterion
Candidate 2 (PK5) vs H_2 Base	1.5868	0.8400	0.1644	Within Criterion
MM vs PK5 (Mutual Agreement)	0.3003	0.2370	0.1420	High Consistency

7.4.2. Reconciliation of Pointwise Force Discrepancies vs Global L_2 Error

Fig. 7.2 plots the pointwise relative force error $\frac{|RF_1 - RF_{1,H_1}|}{RF_{1,H_1}}$ across Domain A. A critical distinction must be made between **global integral error metrics** and **localized pointwise discrepancies**:

- **Global Integral Metrics:** The normalized L_2 curve error (1.4249% for MM, 1.1467% for PK5) and relative external work error (0.8134% for MM, 0.5751% for PK5) integrate across the entire loading interval and lie strictly within the provisional 2.0% accuracy criterion.
- **Pointwise Discrepancies near Initiation:** Near the onset of non-linear softening ($u_1 \approx 8.4\text{--}8.7$ μm), the pointwise relative error curve exhibits a localized transient peak reaching 3.81% for MM (at $u_1 = 8.69$ μm) and 3.21% for PK5 (at $u_1 = 8.68$ μm). The local force-error peak coincides with the difference in saved-frame damage-initiation brackets between the uniform and locally refined discretizations. Once non-linear softening fully engages, the pointwise force difference settles back to $\sim 1.04\%$ (MM) and $\sim 0.70\%$ (PK5).

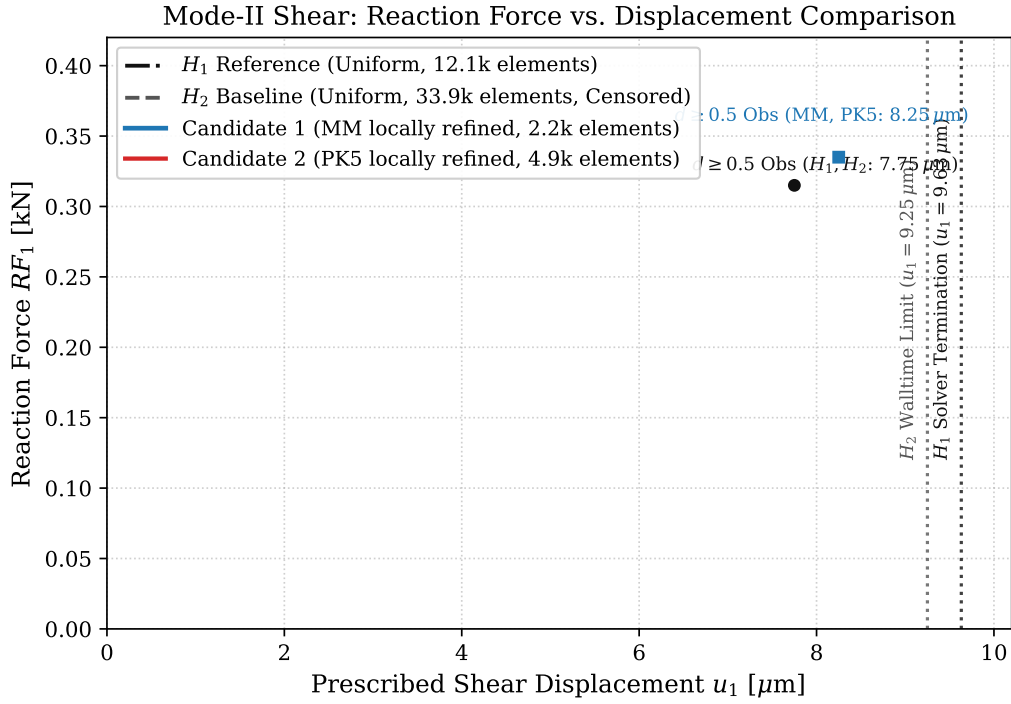


Fig. 7.1.: Mode-II shear reaction force vs. prescribed displacement curves. The H_1 reference terminates at $u_1 = 9.63 \mu\text{m}$, while the H_2 baseline is walltime-censored at $u_1 = 9.25 \mu\text{m}$. Both production locally refined models (MM and PK5) complete the full loading trajectory to $u_1 = 10.0 \mu\text{m}$. Discrete saved-frame damage initiation observations ($d \geq 0.5$) are annotated.

7.4.3. Damage Initiation Observations ($d \geq 0.5$)

Discrete field frame observations record the first occurrence of $d_{\max} \geq 0.5$ at $u_1 = 0.007750 \text{ mm}$ for H_1 ($d_{\max} = 0.5400$) and H_2 ($d_{\max} = 0.5925$), compared to $u_1 = 0.008250 \text{ mm}$ for MM ($d_{\max} = 0.5262$) and PK5 ($d_{\max} = 0.5700$), as summarized in Tab. 7.4.

Tab. 7.4.: Damage Initiation Observations ($d \geq 0.5$) across Saved Field Frames.

Model	Prior Frame (u_1, d)	Crossing Frame (u_1, d)	Continuous Bracket
H_1 Reference	0.007500 mm ($d = 0.4578$)	0.007750 mm ($d = 0.5400$)	$0.00750 < u_1 \leq 0.00775 \text{ mm}$
H_2 Baseline	0.007500 mm ($d = 0.4876$)	0.007750 mm ($d = 0.5925$)	$0.00750 < u_1 \leq 0.00775 \text{ mm}$
Candidate 1 (MM)	0.008000 mm ($d = 0.4543$)	0.008250 mm ($d = 0.5262$)	$0.00800 < u_1 \leq 0.00825 \text{ mm}$
Candidate 2 (PK5)	0.008000 mm ($d = 0.4863$)	0.008250 mm ($d = 0.5700$)	$0.00800 < u_1 \leq 0.00825 \text{ mm}$

7.4.4. Domain-B Continuation Diagnostics ($0.009250 < u_1 \leq 0.010000 \text{ mm}$)

Because neither uniform fine reference (H_1 nor H_2) completed the full post-peak range into Domain B, continuation consistency is assessed as an **internal mutual diagnostic** between the two adaptive sizing strategies.

The mutual L_2 difference between MM and PK5 in Domain B is 0.3402%, with a relative external work discrepancy of 0.3074% and a terminal force discrepancy of 0.64% (0.373693 kN vs 0.371313 kN at $u_1 = 0.01000 \text{ mm}$). This demonstrates high internal consistency during crack propagation.

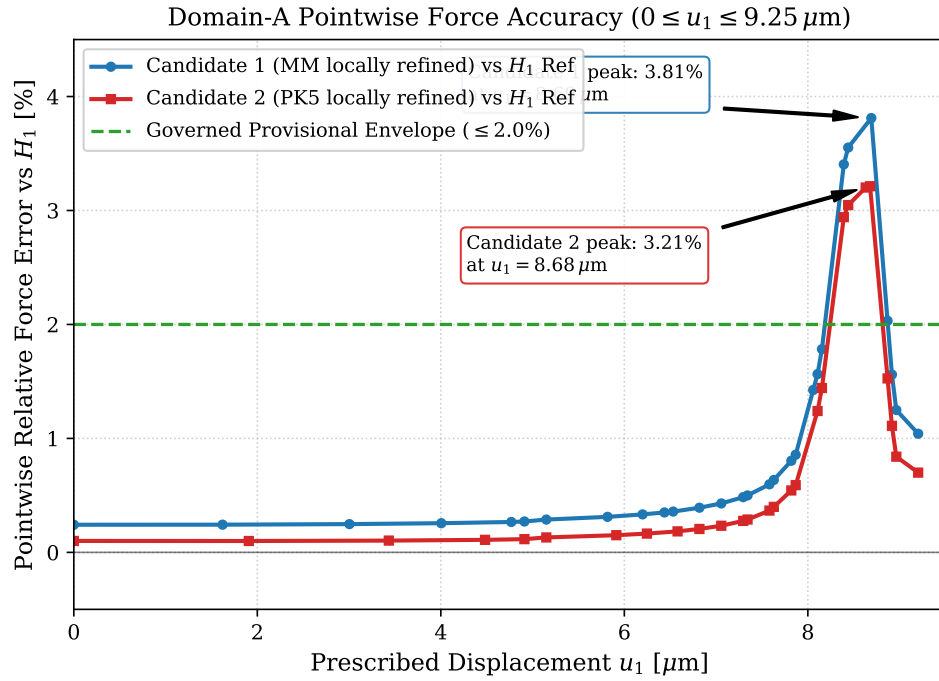


Fig. 7.2.: Pointwise relative force discrepancy of locally refined candidates vs. the authoritative H_1 uniform reference over Domain A ($0 \leq u_1 \leq 9.25 \mu\text{m}$). While global normalized L_2 errors lie strictly below 1.43%, localized pointwise discrepancies temporarily reach peak values of 3.81% (MM) and 3.21% (PK5) near damage initiation ($u_1 \approx 8.4\text{--}8.7 \mu\text{m}$), coinciding with the difference in discrete saved-frame damage-initiation brackets, before settling back to $\sim 1.0\%$ or lower during softening.

7.5. Computational Efficiency and Runtime Scaling

Relative to the uniform ultra-fine H_2 baseline (CPU time = 14,455.0 s, walltime-censored at $u_1 = 0.009250 \text{ mm}$), the locally refined models achieved total CPU execution times of 1,189.0 s (MM, 19.98 min walltime) and 2,609.0 s (PK5, 43.70 min walltime), as illustrated in Fig. 7.3.

These reflect diagnostic scheduler-CPU ratios of **12.16×** (MM) and **5.54×** (PK5) relative to the censored H_2 baseline. Because H_2 stopped prematurely due to queue limits, these comparisons represent diagnostic elapsed-time indicators rather than strict like-for-like full-range speedups.

7.6. Energy Bookkeeping and Methodological Scope

7.6.1. External Work and Internal Dissipation Accounting

- **External Work Tracking:** External work (W_{ext}) is accurately evaluated via reaction force integration ($2.13922 \times 10^{-3} \text{ kN} \cdot \text{mm}$ for MM; $2.13382 \times 10^{-3} \text{ kN} \cdot \text{mm}$ for PK5), showing $< 0.82\%$ error relative to H_1 .
- **UEL Internal Energy Accounting:** Standard Abaqus material energy variables (ALLIE, ALLSE, ALLPD) report zero in standard output tables because all constitutive

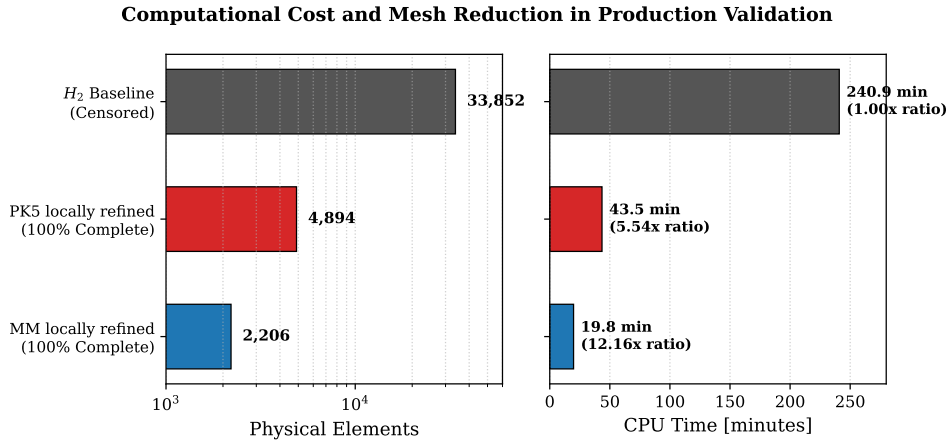


Fig. 7.3.: Mesh size reduction (log scale) and CPU execution time comparison between the fine H_2 baseline (censored at $u_1 = 9.25 \mu\text{m}$) and the complete production locally refined models (MM and PK5).

degradation and fracture dissipation are computed internally within user subroutine `f42_mixed_uel.for`. Consequently, raw `ETOTAL = -ALLWK` in Abaqus output is an expected software accounting artifact and is not interpreted as physical energy non-conservation. A complete UEL-aware global thermodynamic energy balance was not independently available from standard Abaqus output. External work is therefore used as a comparative global-response metric, while phase/history admissibility is assessed independently.

7.6.2. Scope Boundary: Pre-Refinement vs Online Adaptivity

The results demonstrate that fixed, error-guided non-uniform meshes generated prior to fracture simulation achieve high accuracy and substantial computational savings for localized shear fracture. However, this workflow operates on fixed meshes optimized prior to loading; fully automated online remeshing with dynamic mesh updates during crack growth remains a documented future extension.

7.7. Chapter Summary

The production validation confirms the effectiveness and efficiency of error-guided mesh refinement:

1. Both MM (2,206 elements) and PK5 (4,894 elements) achieve tight quantitative agreement with the uniform H_1 reference over Domain A (L_2 errors of 1.42% and 1.15%, stiffness agreement within 0.25%).
2. Discrete frame audits confirm zero violations of phase bounds, history bounds, and temporal irreversibility across all 72 saved ODB states.
3. In comparison to the uniform H_2 baseline, the error-guided meshes yield diagnostic scheduler-CPU ratios of 12.16 $\times$ (MM) and 5.54 $\times$ (PK5) relative to the censored H_2

baseline, making high-resolution phase-field fracture simulation practical on modest computational resources.

8. Synthesis, Decision Framework, and Conclusions

8.1. Synthesis of Validated Research Outcomes

This thesis investigated the implementation, verification, and performance of error-indicator-guided mesh refinement and multi-field state transfer for phase-field brittle fracture simulations in Abaqus. Tab. 8.1 summarizes the core research areas and their validated scientific outcomes.

Tab. 8.1.: Comprehensive Synthesis of Thesis Research Outcomes and Validation Boundaries.

Research Area	Methodology / Implementation	Status	Validated Scientific Boundary
Mode-I Baseline Verification	Molnar staggered UEL/UMAT formulation [18] on SENT benchmark	Validated	Captures global force scale (0.7617 kN) and crack path. Demonstrates 0.11% L_2 convergence between H_1 and H_2 -PUB.
Error-Guided Pre-Refinement	Abaqus MisesERI extraction from elastic pre-analysis [21]	Validated	Recovers Mode-I peak force ($< 0.25\%$ error) with 14.7% fewer elements and 30.7% CPU time reduction vs H_1 .
Multi-Field State Transfer	Inverse isoparametric projection for $(\mathbf{u}, d)$ and history clamping ($\mathcal{H} \geq 0$)	Validated	Demonstrated across same-mesh restarts, multi-resolution refined/coarsened grids, and four-stage restart protocols.
Active-Set Admissibility	KKT multiplier analysis on frozen active damage sets	Validated	Demonstrates that a fixed active set held frozen without dynamic updates is inadmissible under advancing load and supports the history-field formulation adopted in the present restart framework.
Synthetic Topology Transfer	Single-facet crack-tip extension benchmark with slit isolation	Validated	Achieves 194 accepted post-peak continuation increments with $> 74.5\%$ load drop while preserving continuum bounds.
Production Mode-II Validation	Error-guided MM (2,206 elem) and PK5 (4,894 elem) models vs H_1/H_2	Validated	Domain-A L_2 errors $< 1.43\%$, work errors $< 0.82\%$, stiffness within 0.25%. Diagnostic scheduler-CPU ratios of 12.16 $\times$ and 5.54 $\times$ relative to censored H_2 .

8.2. Engineering Decision Tree for Phase-Field Mesh Adaptation

Based on the quantitative evidence established across this work, a practical engineering decision framework for selecting mesh refinement and state-transfer strategies in phase-field

fracture simulations is established:

1. Initial Stress Concentration Assessment:

- If the prospective crack initiation and propagation corridor is localized and predictable from the initial elastic stress field (as in notched specimens or stress concentrators), utilize **Offline Error-Guided Pre-Refinement** (the Pandey–Kumar approach [21]).
- If maximum computational throughput is desired, apply a **Minimum-Maximum sizing rule** (MM, bounds [1.0%, 5.0%]), yielding a 12.16× diagnostic scheduler-CPU ratio relative to the censored H_2 baseline with $< 1.43\%L_2$ error.
- If enhanced resolution throughout the process zone is required, apply a **Uniform-Error sizing rule** (PK5, target 5.0%), yielding a 5.54× diagnostic scheduler-CPU ratio with $< 1.15\%L_2$ error.

2. Evolving or Unpredictable Crack Paths:

- When crack paths turn, bifurcate, or develop far from initial stress concentrators, multi-stage mesh update and state transfer are necessary.
- For the present validated implementation, state transfer uses the **Four-Stage Restart Protocol** (STATE_INSTALL $\rightarrow$ MECH_EQUILIBRATION with phase locked $\rightarrow$ PHASE_RELEASE $\rightarrow$ CONTINUATION) to eliminate the observed startup transients.
- Slit-barrier topological isolation must be enforced along existing crack flanks to prevent unphysical cross-slit smoothing.
- History-field tracking ($\mathcal{H} \geq 0$) must be employed across restart boundaries rather than static frozen active sets.

8.3. Methodological and Scientific Limitations

To maintain rigorous scientific discipline, the following limitations are explicitly documented:

1. **Pre-Refinement vs Runtime Adaptivity:** The validated production models operate on fixed non-uniform meshes generated prior to loading. While highly effective for stress-concentration-driven fracture, this approach cannot dynamically adapt to unexpected crack turning, branching, or secondary micro-cracking far from the initial elastic concentration zone.
2. **Censored Uniform Post-Peak References:** In the Mode-II shear benchmark, fine uniform meshes (H_1 and H_2) terminated prematurely due to solver cutbacks and queue walltime limits. Consequently, post-peak validation in Domain B relies on internal mutual consistency between adaptive strategies rather than an uncensored uniform fine-mesh reference.

3. **Spatial Crack Path Mesh Sensitivity:** Matched-state Hausdorff distances between H_1 and H_2 ($5.44 \mu\text{m}$, exceeding the half-element threshold $h_{H2}/2 = 3.75 \mu\text{m}$ for the fine $h_{H2} = 7.5 \mu\text{m}$ grid) demonstrate that spatial fracture geometry remains sensitive to mesh resolution ($h/l_0 = 0.5$ vs 1.0) even when global force response exhibits tight convergence (0.518%).
4. **UEL Energy Bookkeeping:** Standard Abaqus material energy variables report zero because constitutive degradation is evaluated inside custom user subroutines. A complete UEL-aware global thermodynamic energy balance was not independently available from standard Abaqus output. External work is therefore used as a comparative global-response metric, while phase/history admissibility is assessed independently.

8.4. Future Research Roadmap

The methodologies established in this thesis provide a strong foundation for future research:

- **Automated In-Analysis Dynamic Remeshing:** Implementing an automated driver that monitors phase-field gradients $|\nabla d|$ during solver increments, triggers runtime remeshing when damage approaches unrefined zones, and automatically transfers state using the validated four-stage protocol.
- **Three-Dimensional Implementation:** Extending the multi-field transfer operators and layered UEL architecture to 3D hexahedral/tetrahedral elements with octree-based spatial search.
- **Distributed-Memory (MPI) Parallelization:** Developing domain-decomposition state-transfer algorithms to enable massively parallel phase-field simulations on distributed HPC clusters.

8.5. Conclusions

This thesis demonstrated that error-indicator-guided mesh refinement and multi-field state transfer provide a practical, highly efficient framework for phase-field fracture modeling in commercial finite element software. By leveraging Abaqus built-in MISESeri error indicators during an elastic pre-analysis, non-uniform meshes achieve approximately twelve-fold lower scheduler CPU time than the censored H_2 run while strictly maintaining global force-displacement accuracy ($< 1.43\% L_2$ error), initial elastic stiffness ($< 0.25\%$ error), and continuum state admissibility. The multi-field transfer algorithms, four-stage restart protocol, and decision framework established herein provide a rigorous basis for future adaptive fracture mechanics workflows.

A. Reproducibility Tables, Job Execution Provenance, and Diagnostic Logs

A.1. Master High-Performance Computing Job Ledger

To ensure complete scientific reproducibility, Tab. A.1 provides the comprehensive record of all authoritative HPC simulations executed across the research campaign on the TU Bergakademie Freiberg cluster.

Tab. A.1.: Authoritative HPC Simulation Job Ledger across All Research Stages.

Stage	Job Name	PBS ID	CPUs	RAM	Outcome / Role
Stage A	SingleNotch_v2	1374864	1	32 GB	Baseline Molnar SENT verification (Exit 0, 00:38:38)
Stage A	M2REF_H0	1386372	1	8 GB	Uniform coarse baseline, $h/l_0 = 2.0$ (Exit 0, 00:33:24)
Stage A	M2REF_H1	1386447	1	8 GB	Uniform refined baseline, $h/l_0 = 1.0$ (Exit 1, 01:30:53)
Stage A	M2REF_H2	1386448	1	8 GB	Uniform fine diagnostic, $h/l_0 = 0.5$ (Exit -29, 04:01:41)
Stage C	F43PRE3_GEOM	1385461	1	8 GB	Elastic MISESRI pre-analysis (Exit 0, 00:01:15)
Stage C	F43REM4_MM	1385575	1	8 GB	Min-Max remeshed input generation (Exit 0, 00:02:10)
Stage C	F43REM4_PK5	1385574	1	8 GB	Uniform-Error remeshed generation (Exit 0, 00:03:45)
Stage D	D2A_INGEST	1388706	1	8 GB	Multi-field binary state ingestion (Exit 0, 00:08:12)
Stage D	D3A3_RESTART	1389718	1	16 GB	Native restart reproducibility (0.0001%, Exit 0)
Stage D	D3D_ACTIVESET	1389715	1	16 GB	Active-set multiplier violation proof (Exit 0)
Stage E	M2_REF_TRANSFER	1390829	1	16 GB	Refined target grid transfer verification (Exit 0)
Stage E	M2_CRS_TRANSFER	1390830	1	16 GB	Coarsened target grid transfer verification (Exit 0)
Stage F	M2STAGE_F_VAL	1393159	1	16 GB	Synthetic topology transfer, 194 incs (Exit 1)
Stage G	M2PROD_ADAPT_MM	1394260	1	16 GB	Production Min-Max locally refined run (Exit 0; CPU 00:19:49; wall 00:19:59)
Stage G	M2PROD_ADAPT_PK5	1394261	1	16 GB	Production Uniform-Error locally refined (Exit 0; CPU 00:43:29; wall 00:43:42)

A.2. Mode-I Uniform-Convergence Provenance

The Mode-I $H_0/H_1/H_2$ -PUB values in Tab. 2.2 are traceable to the frozen Molnar $l_0 = 0.015$ mm study family:

Consolidated CAE job 1376236.mmast02 generated the three archived histories under `results/processed/molnar_lc015_h_convergence/source_csv`. The governing scientific-input revision is 58d7e3102d76fe0e70e6729457e2c7e90ad131bb. Immutable deck, source, and response-history hashes are recorded in the frozen provenance record `MOLNAR_MESH_ROLE_AND_RESULT_FREEZE.md`.

Tab. A.2.: Traceability of the Mode-I Uniform-Convergence Data.

Case	Solver Job	Generated Deck	Archived RF2–U2 History
H_0	1376154	H0_exact/SingleNotch.inp	H0_RF2_U2.csv
H_1	1376185	H1_h0025/H1_h0025.inp	H1_RF2_U2.csv
H_2 -PUB	1376186	H2_pub_h0010/H2_pub_h0010.inp	H2-PUB_RF2_U2.csv

A.3. Cryptographic Checksums and Package Provenance

Tab. A.3 records the exact SHA-256 cryptographic hashes for the primary input decks, Fortran user subroutines, PBS directives, and manifests executed in the production analyses.

Tab. A.3.: Cryptographic SHA-256 Checksums for Production Executable Artifacts (Input Deck, User Subroutine, PBS Directives, and Package Manifest).

Model / Artifact	Component	Exact SHA-256 Cryptographic Hash
M2PROD_ADAPT_MM	Input Deck (.inp)	5b8cb39b08d594953d884212167b8a9578349cc397352e52f253dedd819d326d
	User Subroutine (.for)	0bc4378179a35acd9954d20d3e07517f8e1c356ae07a23c40e7715cd7b56dce8
	PBS Script (.pbs)	678fcae0c7ae48b120d61bbd255ead5538388aed39526e3f0c34aa45c29a07db
	Manifest (.json)	d9ae32c3cba00e8673475f1b4e9da56bcfcf2fa890243ad649d584422eb02381
M2PROD_ADAPT_PK5	Input Deck (.inp)	402f6b61f59a31bd1ebee77019e47a3a7da0557296660c20ddc8aaab30b847f4
	User Subroutine (.for)	0bc4378179a35acd9954d20d3e07517f8e1c356ae07a23c40e7715cd7b56dce8
	PBS Script (.pbs)	7ad83eff8ba7b46e69e924ad5218e81d187e9d0193521cfff9f446ab662a176f9
	Manifest (.json)	b069232b2db66dacf537329ec3be9a0a1a39d6b5bfff6f6a18e1097da2706c52
M2STAGE_F_VAL	Input Deck (.inp)	168f4fd6777c3e7c110747b800c0565fa93f16fec32e655adf67f2cb3ef470aa
	User Subroutine (.for)	62e35f74bbeccd3f5b1ac67312b79211f4ccb75648ae20dbb8985cb577fe1aab
	Committed Binary	c63681b1c0b029f7b3bf5a1ee986e3fabfbdb240cf0bc89a030a811b82a84722

A.4. Historical Solver Diagnostic Codes and Failure Mode Catalog

To document the technical challenges resolved during the research campaign, Tab. A.4 provides a diagnostic catalog of key software failure modes encountered and their architectural resolutions.

A.5. Operational HPC Monitoring Protocols

To ensure complete lifecycle tracking during long-running cluster executions, a dual-channel monitoring protocol was employed:

- **PBS System Mail:** Automated notifications via #PBS -m a be on job begin, abort, and normal termination.
- **Webhook Notifications:** Auxiliary completion hooks logging scheduler exit codes, CPU runtimes, and output file sizes, with credentials isolated in secure local configuration files (chmod 600) outside the repository.

Tab. A.4.: Catalog of Technical Failure Modes and Engineering Resolutions.

Error Identifier	Observed Manifestation	Root Cause Diagnosis	Architectural Resolution
ErrElemAreaSmallNegZero	Abaqus pre-solver aborts with zero/negative area elements	Global node number collision between auxiliary Reference Point and mesh vertices	Strict *Part node scoping isolating RP node from mesh assembly
FortranImplicitReal8	Compiler typing error in INQUIRE statement	Implicit type assignment of EXISTS as REAL*8 instead of LOGICAL	Explicit type declarations (LOGICAL :: file_exists) across all sub-routines
ModuleCompilerMissing	ifort: command not found on compute nodes	PBS execution environment omitted required toolchain modules	Explicit module load intel/2024.2.0 gcc/11.4.0 in all job wrappers
AbaqusPy2UnicodeError	TypeError in RemeshingRule.variables	Abaqus Python 2.7 passed Unicode string u'MISESeri' instead of native string	Explicit ASCII string conversion str('MISESeri') in CAE generator scripts
CrossSlitContamination	Artificial phase smoothing across physical notch flanks	Nonmatching geometric search interpolated across opposite slit surfaces	Slit-barrier topological partitioning restricting search to same flank

Bibliography

- [1] Ambati, M., Gerasimov, T., and De Lorenzis, L. A review on phase-field models of brittle fracture and a new fast hybrid formulation. *Computational Mechanics*, 55(2): 383–405, 2015. doi:[10.1007/s00466-014-1109-y](https://doi.org/10.1007/s00466-014-1109-y).
- [2] Badnava, H., Borden, M. J., and Rabczuk, T. Adaptive mesh refinement for phase field fracture using recovery based error estimation. *Computer Methods in Applied Mechanics and Engineering*, 367:113098, 2020. doi:[10.1016/j.cma.2020.113098](https://doi.org/10.1016/j.cma.2020.113098).
- [3] Belytschko, T., Liu, W. K., Moran, B., and Elkhodary, K. *Nonlinear Finite Elements for Continua and Structures*. John Wiley & Sons, Chichester, UK, 2nd edition, 2014.
- [4] Bourdin, B., Francfort, G. A., and Marigo, J.-J. Numerical experiments in revisited brittle fracture. *Journal of the Mechanics and Physics of Solids*, 48(4):797–826, 2000. doi:[10.1016/S0022-5096\(99\)00028-9](https://doi.org/10.1016/S0022-5096(99)00028-9).
- [5] Bourdin, B., Francfort, G. A., and Marigo, J.-J. The variational approach to fracture. *Journal of Elasticity*, 91(1–3):5–148, 2008. doi:[10.1007/s10659-007-9107-3](https://doi.org/10.1007/s10659-007-9107-3).
- [6] Cebal, J. R. and Löhner, R. Conservative load projection and tracking for fluid-structure interaction problems. *AIAA Journal*, 35(4):687–692, 1997. doi:[10.2514/2.172](https://doi.org/10.2514/2.172).
- [7] *Abaqus 2023 Documentation: Analysis User's Guide, User Subroutines Reference Guide, and CAE User's Guide*. Dassault Systèmes Simulia Corp., Providence, RI, USA, 2023.
- [8] Diddige, V., Roth, S., and Kiefer, B. Phase-field modeling of hydrogen-promoted fracture: Natural incorporation of hydrostatic stress dependencies via a chemical potential-based variational formulation. *Computer Methods in Applied Mechanics and Engineering*, 445:118143, 2025. doi:[10.1016/j.cma.2025.118143](https://doi.org/10.1016/j.cma.2025.118143).
- [9] Farhat, C., Lesoinne, M., and Le Tallec, P. Load and motion transfer algorithms for fluid/structure interaction problems with non-matching discrete interfaces. *Computer Methods in Applied Mechanics and Engineering*, 157(1–2):95–114, 1998. doi:[10.1016/S0045-7825\(97\)00234-7](https://doi.org/10.1016/S0045-7825(97)00234-7).
- [10] Francfort, G. A. and Marigo, J.-J. Revisiting brittle fracture as an energy minimization problem. *Journal of the Mechanics and Physics of Solids*, 46(8):1319–1342, 1998. doi:[10.1016/S0022-5096\(98\)00034-9](https://doi.org/10.1016/S0022-5096(98)00034-9).
- [11] Griffith, A. A. The phenomena of rupture and flow in solids. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, 221:163–198, 1921. doi:[10.1098/rsta.1921.0006](https://doi.org/10.1098/rsta.1921.0006).

- [12] Heister, T., Wheeler, M. F., and Wick, T. A primal-dual active set method and predictor-corrector mesh adaptivity for computing fracture propagation using a phase-field approach. *Computer Methods in Applied Mechanics and Engineering*, 290:466–495, 2015. doi:[10.1016/j.cma.2015.03.009](https://doi.org/10.1016/j.cma.2015.03.009).
- [13] Irwin, G. R. Analysis of stresses and strains near the end of a crack traversing a plate. *Journal of Applied Mechanics*, 24(3):361–364, 1957. doi:[10.1115/1.4011547](https://doi.org/10.1115/1.4011547).
- [14] Jiao, X. and Heath, M. T. Common-refinement-based data transfer between non-matching meshes in multiphysics simulations. *International Journal for Numerical Methods in Engineering*, 61(14):2402–2427, 2004. doi:[10.1002/nme.1147](https://doi.org/10.1002/nme.1147).
- [15] Mesgarnejad, A., Bourdin, B., and Khonsari, M. M. Validation simulations for the variational approach to fracture. *Computer Methods in Applied Mechanics and Engineering*, 290:420–437, 2015. doi:[10.1016/j.cma.2015.02.019](https://doi.org/10.1016/j.cma.2015.02.019).
- [16] Miehe, C., Hofacker, M., and Welschinger, F. A phase field model for rate-independent crack propagation: Robust algorithmic implementation based on operator splits. *Computer Methods in Applied Mechanics and Engineering*, 199(45–48):2765–2778, 2010. doi:[10.1016/j.cma.2010.04.011](https://doi.org/10.1016/j.cma.2010.04.011).
- [17] Miehe, C., Welschinger, F., and Hofacker, M. Thermodynamically consistent phase-field models of fracture: Variational principles and multi-field FE implementations. *International Journal for Numerical Methods in Engineering*, 83(10):1273–1311, 2010. doi:[10.1002/nme.2861](https://doi.org/10.1002/nme.2861).
- [18] Molnar, G. and Gravouil, A. 2D and 3D Abaqus implementation of a robust staggered phase-field solution for modeling brittle fracture. *Finite Elements in Analysis and Design*, 130:27–38, 2017. doi:[10.1016/j.finel.2017.03.002](https://doi.org/10.1016/j.finel.2017.03.002).
- [19] Msekh, M. A., Sargado, J. M., Jamshidian, M., Areias, P. M., and Rabczuk, T. Abaqus implementation of phase-field model for brittle fracture. *Computational Materials Science*, 96:472–484, 2015. doi:[10.1016/j.commatsci.2014.05.071](https://doi.org/10.1016/j.commatsci.2014.05.071).
- [20] Nagaraja, S., Elhaddad, M., Ambati, M., and De Lorenzis, L. A dynamic-local adaptive mesh refinement strategy for phase-field modeling of fracture. *International Journal for Numerical Methods in Engineering*, 118(13):783–809, 2019. doi:[10.1002/nme.6036](https://doi.org/10.1002/nme.6036).
- [21] Pandey, A. and Kumar, S. A simple and robust mesh refinement implementation in Abaqus for phase field modelling of brittle fracture. *Computer Modeling in Engineering & Sciences*, 144(3):3251–3286, 2025. doi:[10.32604/cmescs.2025.060124](https://doi.org/10.32604/cmescs.2025.060124).
- [22] Paul, K., Zimmermann, C., Kuhn, C., and Müller, R. Adaptive mesh refinement in phase-field fracture simulations using hierarchical B-splines and quadrilateral meshes. *Computational Mechanics*, 66(4):835–854, 2020. doi:[10.1007/s00466-020-01878-5](https://doi.org/10.1007/s00466-020-01878-5).
- [23] Perić, D., Hochard, C., Dutko, M., and Owen, D. R. J. Transfer operators for evolving meshes in small strain elasto-plasticity. *Computer Methods in Applied Mechanics and Engineering*, 137(3–4):331–344, 1996. doi:[10.1016/0045-7825\(96\)01018-0](https://doi.org/10.1016/0045-7825(96)01018-0).
- [24] Zienkiewicz, O. C. and Zhu, J. Z. A simple error estimator and adaptivity for practical engineering analysis. *International Journal for Numerical Methods in Engineering*, 24(2):337–357, 1987. doi:[10.1002/nme.1620240206](https://doi.org/10.1002/nme.1620240206).

-
- [25] Zienkiewicz, O. C. and Zhu, J. Z. The superconvergent patch recovery and a posteriori error estimates. Part 1: The recovery technique. *International Journal for Numerical Methods in Engineering*, 33(7):1331–1364, 1992. doi:[10.1002/nme.1620330702](https://doi.org/10.1002/nme.1620330702).